Weighbridge Management System

Product Manual

Version 1.2.0 · Truck weighing, gate entry, plate recognition, invoice OCR with weighbridge matching, and SAP / Oracle integration

Prepared for	
Site	
Installed by	
Date	

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1. Introduction

The **Weighbridge Management System** records every truck that is weighed on your weighbridge(s), controls the gate barriers, and passes the completed records to your Oracle database. It replaces manual registers and paper slips with a single screen that the operator can learn in minutes.

What it does

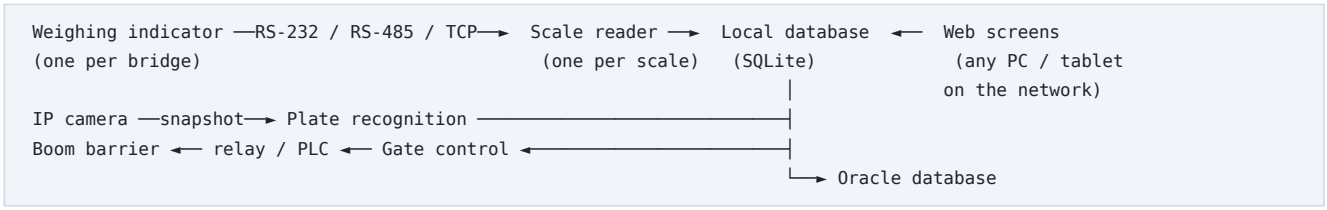
Area	Capability
Weighing	Reads the weight directly from the indicator (RS-232, RS-485, TCP or Modbus), accepts it only when **stable**, prints a weighment slip, keeps a permanent record. Two-pass (gross then tare, or tare then gross) or single-pass using a stored vehicle tare.
Several bridges	Any number of scales on one PC, each with its own port, settings and camera. A truck may be weighed in on one bridge and out on another.
Plate recognition	Optional. Reads the vehicle number from a camera, fills the form, and checks that the truck on the bridge is the truck on the ticket.
Gate entry	Register vehicles at the gate, print a gate pass, keep a live list of vehicles inside, block problem vehicles, control the exit.
Gate barriers	OPEN / CLOSE buttons for every boom barrier; barriers can open automatically when a weighing completes.
Invoice documents	Upload a picture or PDF of a purchase invoice or a material-return document; the system reads it (OCR), you check it, and it is **matched with the weighbridge tickets**. Differences (wrong quantity, wrong truck, no ticket, double billing, over PO quantity…) are listed as **exceptions**. Verified documents are sent to **SAP** and/or **Oracle**.
Oracle / SAP	Completed tickets are sent to your Oracle table automatically, safely and without duplicates, even if the network is temporarily down.
Control	Users and roles, audit trail, reports with CSV export, backups, a read-only interface for ERP systems.

Important limits

- The system records what the **indicator** sends. If the weights are used for commercial billing, the indicator and weighbridge must be legally approved and sealed for trade in your country. This software does not replace calibration or statutory certification.
- The software sends only OPEN/CLOSE contacts to a barrier. **Anti-crush safety** (safety loop, photocell, limit switches, emergency stop) must be provided by the barrier hardware. See section 8.

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2. System overview



The program runs on **one PC (the "weighbridge server")** next to the weighbridge. Operators use it from a web browser — on that PC, or from any other PC, laptop or tablet on the network. There are two background programs:

- **Scale supervisor** – starts a reader for every enabled scale, restarts it if it stops, and closes gates after their auto-close time.
- **Oracle sync** *(optional)* – sends completed tickets, and verified documents (to Oracle and/or SAP), every 30 seconds.

Screens

Screen	Who	Purpose
Weighing	Operator	Live weight of each scale, create tickets, second weighment
Gate	Security	Gate OPEN/CLOSE, register vehicles, vehicles inside, exit
Documents	Everyone	Upload invoices / return documents, review, verify, send
Exceptions	Everyone	Differences between documents and the weighbridge; weighed-but-no-invoice
Reports	Everyone	Find tickets, totals, CSV, Oracle status
Masters	Everyone (delete/block: admin)	Vehicles (with stored tare), parties, materials
Setup	Administrator	Scales, gates, plate recognition, Oracle, company details, users, diagnostics

Roles

Can do	Operator	Administrator
Weigh, print slips, register vehicles, open/close gates, view reports	✔	✔
Add vehicles, parties, materials	✔	✔
Cancel a ticket or gate entry, delete master data, block vehicles		✔
Override a blocked vehicle, plate mismatch or exit rule (reason is recorded)		✔
Upload, read, verify documents; link tickets; mark exceptions resolved (with a note)	✔	✔
Waive an exception, cancel a document, send a held document anyway		✔
Setup page, users, send to Oracle now		✔

3. What you need

3.1 Computer

Item	Requirement
PC	Any modern PC or industrial PC, 4 GB RAM, 10 GB free disk. Windows 10/11, Windows Server 2016+, or Linux (Ubuntu 20.04+ / Debian 11+).
Runs 24×7?	Recommended: yes, with a UPS. The PC must be on for weighing and gate control.
Network	Needed for other PCs/tablets, cameras, network relays and Oracle.

3.2 Software (installed by you or the integrator)

Software	Notes
PHP 8.1 or newer	With extensions `pdo_sqlite`, `sodium`, `mbstring` (required) and `curl` (needed for cameras, plate recognition and HTTP gate relays).
Web browser	Chrome, Edge or Firefox (current).
Oracle client *(optional)*	PHP `oci8` extension and Oracle Instant Client, only for Oracle integration.
Plate recognition service *(optional)*	CodeProject.AI Server, Plate Recognizer, or OpenALPR.
Invoice reader *(optional, one of)*	An **Anthropic API key** (Claude reads photos and PDFs; needs internet, documents are sent to Anthropic) **or Tesseract OCR** installed on the PC (offline; best for clean scans). Without either, documents are typed in by hand.
PHP extensions for documents *(recommended)*	`gd` and `exif` (turns phone photos upright and resizes them), `curl`.

There is nothing else to install: no separate database server, no Composer, no Node.js.

3.3 Field hardware

Item	Notes
Weighing indicator	Must have a serial output (RS-232 or RS-485) or a network gateway. Note its baud rate, data bits, parity and stop bits (typically 9600, 8, N, 1).
Cable / adapter	RS-232: TX, RX, GND (DB9 pins 2, 3, 5). USB-RS232 or USB-RS485 adapter if the PC has no COM port. RS-485: two wires A and B (D+ and D−) plus a 120 Ω terminator on long runs.
Serial-to-Ethernet gateway *(optional)*	For distances >15 m (RS-232) or when the PC is far from the bridge.
IP camera *(optional)*	Must provide a JPEG **snapshot URL**. One per scale and/or gate.
Boom barrier controller and relay *(optional)*	A dry-contact relay board or PLC output that can be wired to the barrier's OPEN/CLOSE inputs. IP relays, USB relay boards and Modbus relay modules are supported.

4. Installation

Follow **either** 4.1 (Windows) or 4.2 (Linux). Then continue with section 5.

4.1 Windows

1. **Install PHP.** Download *PHP 8.2 or newer, Thread Safe, x64* (zip) from windows.php.net and unzip to `C:\php`.
2. **Enable extensions.** Copy `C:\php\php.ini-production` to `C:\php\php.ini`. Open it in Notepad and make sure these lines are present without a leading semicolon: `extension_dir = "ext"` `extension=pdo_sqlite` `extension=sodium` `extension=mbstring` `extension=curl` `extension=gd` `extension=exif` Also set the upload limits so phone photos of invoices are accepted (PHP's default of 2 MB is too small): `upload_max_filesize = 12M` `post_max_size = 16M` Add `C:\php` to the system **PATH** (Windows Settings → search "environment variables").
3. **Copy the program.** Copy the whole `weighbridge` folder to `C:\weighbridge`.
4. **Check the PC.** Open *Command Prompt* and run: `cd C:\weighbridge` `php bin\check.php` Every line must show `[OK]` (or a `[WARN]` you understand — for example "Oracle driver not installed" if you do not use Oracle). Fix any `[FAIL]` first.
5. **Quick test.** Double-click **`start.bat`**. Two windows open (readers and web server). Open `http://localhost:8080/` in the browser. Continue with section 5.
6. **Make it permanent (services).** Install *NSSM* (nssm.cc), edit the two paths at the top of `deploy\windows-install.bat`, and run it **as Administrator**. It creates three Windows services that start with the PC: *WeighbridgeScale* (readers), *WeighbridgeOracleSync* (Oracle transfer) and *WeighbridgeWeb* (screens). Close the `start.bat` windows first.
7. **Windows Firewall.** Allow inbound TCP port 8080 if other PCs will connect.

4.2 Linux (Ubuntu / Debian)

```
sudo apt update
sudo apt install -y php-cli php-sqlite3 php-mbstring php-curl php-gd nginx php-fpm   # or apache2 + libapache2-mod-php
sudo mkdir -p /var/www && sudo cp -r weighbridge /var/www/weighbridge
sudo chown -R www-data:www-data /var/www/weighbridge/data
sudo chmod 750 /var/www/weighbridge/data
sudo usermod -aG dialout www-data            # permission to use serial ports
cd /var/www/weighbridge && sudo -u www-data php bin/check.php
```

Quick test: `./start.sh` (opens the web server on port 8080 and starts the readers).

Permanent:

```
sudo cp deploy/*.service /etc/systemd/system/
sudo systemctl daemon-reload
sudo systemctl enable --now weighbridge-scale weighbridge-oracle-sync
```

and publish `public/` with nginx/Apache. Example nginx site:

```
server {
    listen 80; server_name weighbridge.local;
    root /var/www/weighbridge/public; index index.php;
    location / { try_files $uri $uri/ =404; }
    location ~ \.php$ { include snippets/fastcgi-php.conf; fastcgi_pass unix:/run/php/php8.2-fpm.sock; }
    location ~ /\. { deny all; }
}
```

Important: the web server must point at the **`public`** folder only. The `data` folder (database and key) must never be reachable from the web.

4.3 Folder contents

Path	Contents
`public/`	The screens (the only folder the web server publishes)
`src/`, `bin/`	Program code and background programs
`data/`	**Your data**: `weighbridge.sqlite` (database), `app.key` (encryption key), `snapshots/` (camera photos), `logs/` (reader logs), `documents/` (uploaded invoice pictures and PDFs), `backups/`
`sql/oracle_schema.sql`	Oracle table script

Path	Contents
`deploy/`	Service files for Linux and Windows
`docs/`	This manual, screenshots, comparison notes
`tests/`	Automatic self-tests
`VERSION`, `CHANGELOG.md`	Version information

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5. First-time setup (step by step)

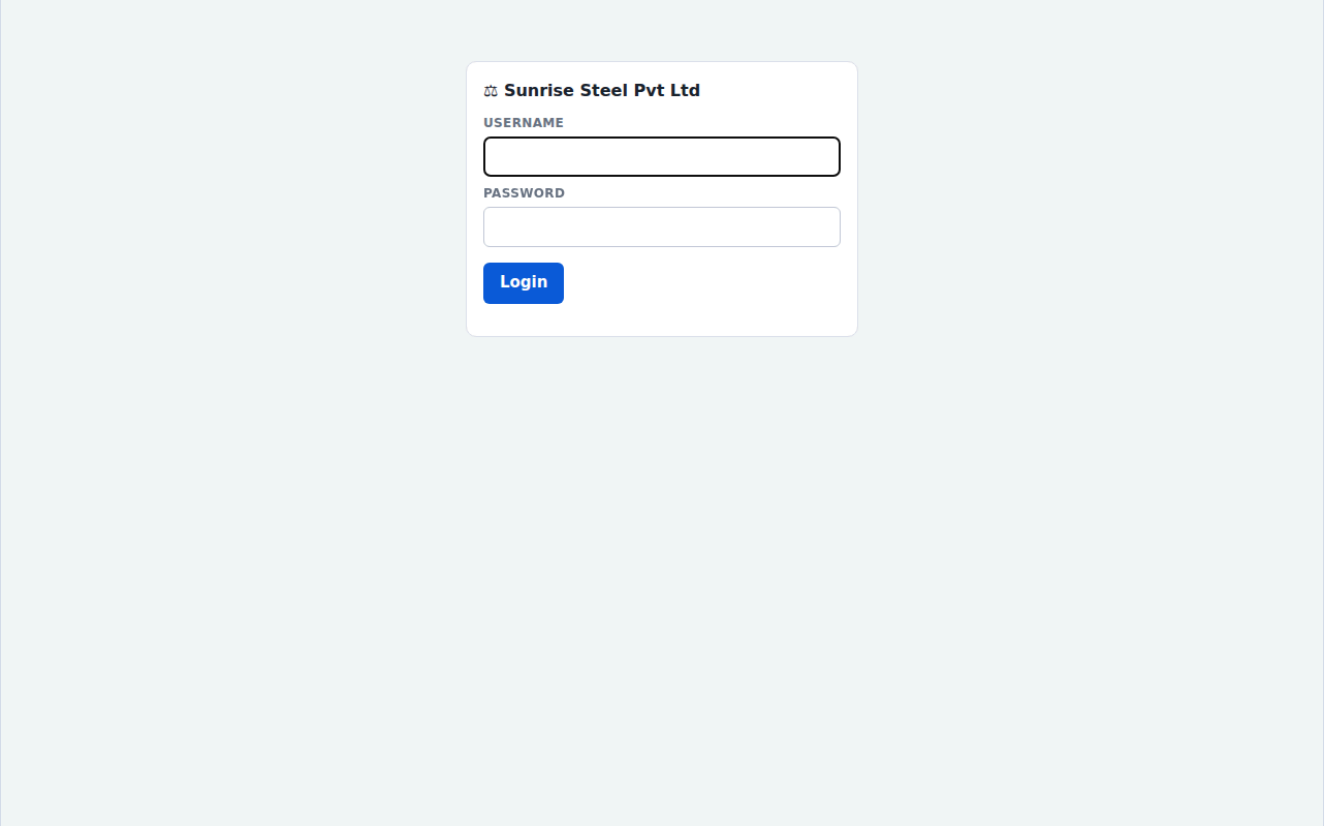
Step 1 — Installer page

Open `http://<server>:8080/install.php`. The page checks the PC (PHP version and extensions, writable data folder). Enter:

- **Company name** (printed on slips)
- **Administrator username and password** (at least 8 characters — keep it safe)

Press **Install**. The system starts with one **Simulator** scale, so you can try everything before connecting real hardware.

Step 2 — Log in



Step 3 — Company details

Setup → General: company name, address (printed on the ticket), ticket prefix (default `WB`, giving numbers such as `WB202609290001`). Tick **Allow manual weight entry** only if you want operators to be able to type a weight when a scale is down (it is flagged on the slip).

Step 4 — Users

Setup → Users: add one **Operator** login per person (never share the administrator login). Minimum password length is 8.

Step 5 — Connect the real scale(s)

See section 6. Switch the scale from *Simulator* to the real port, test, and save.

Step 6 — Masters

Masters: add your regular **vehicles** (with a **stored tare** if you use single-pass weighing), **parties** and **materials**. New names typed on the weighing or gate screens are added automatically.

Step 7 — Optional modules

- Invoice pictures, matching and exceptions: section 13
- Cameras / plate recognition: section 7
- Gates and barriers: section 8
- Oracle: section 9

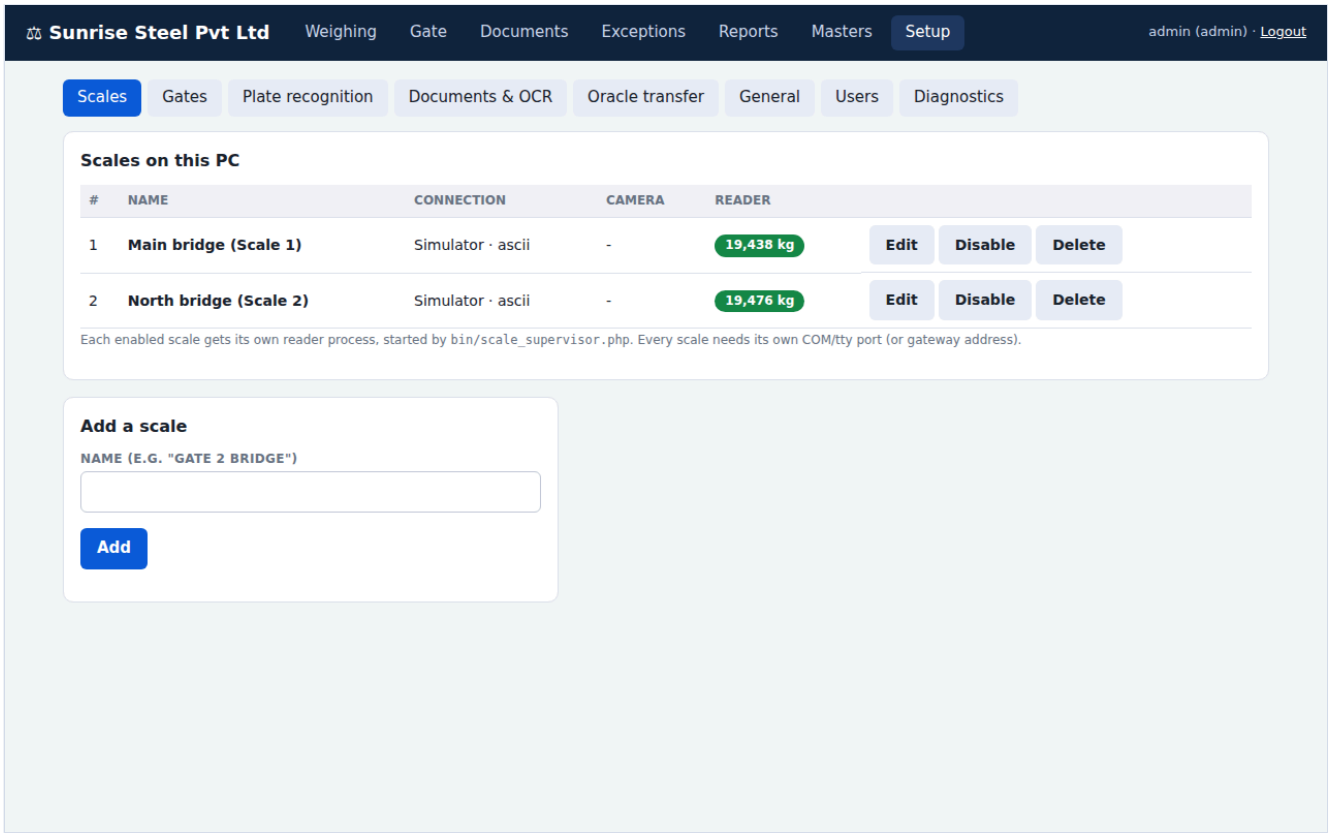
Step 8 — Acceptance test

Complete the checklist in **Appendix D** before handing over.

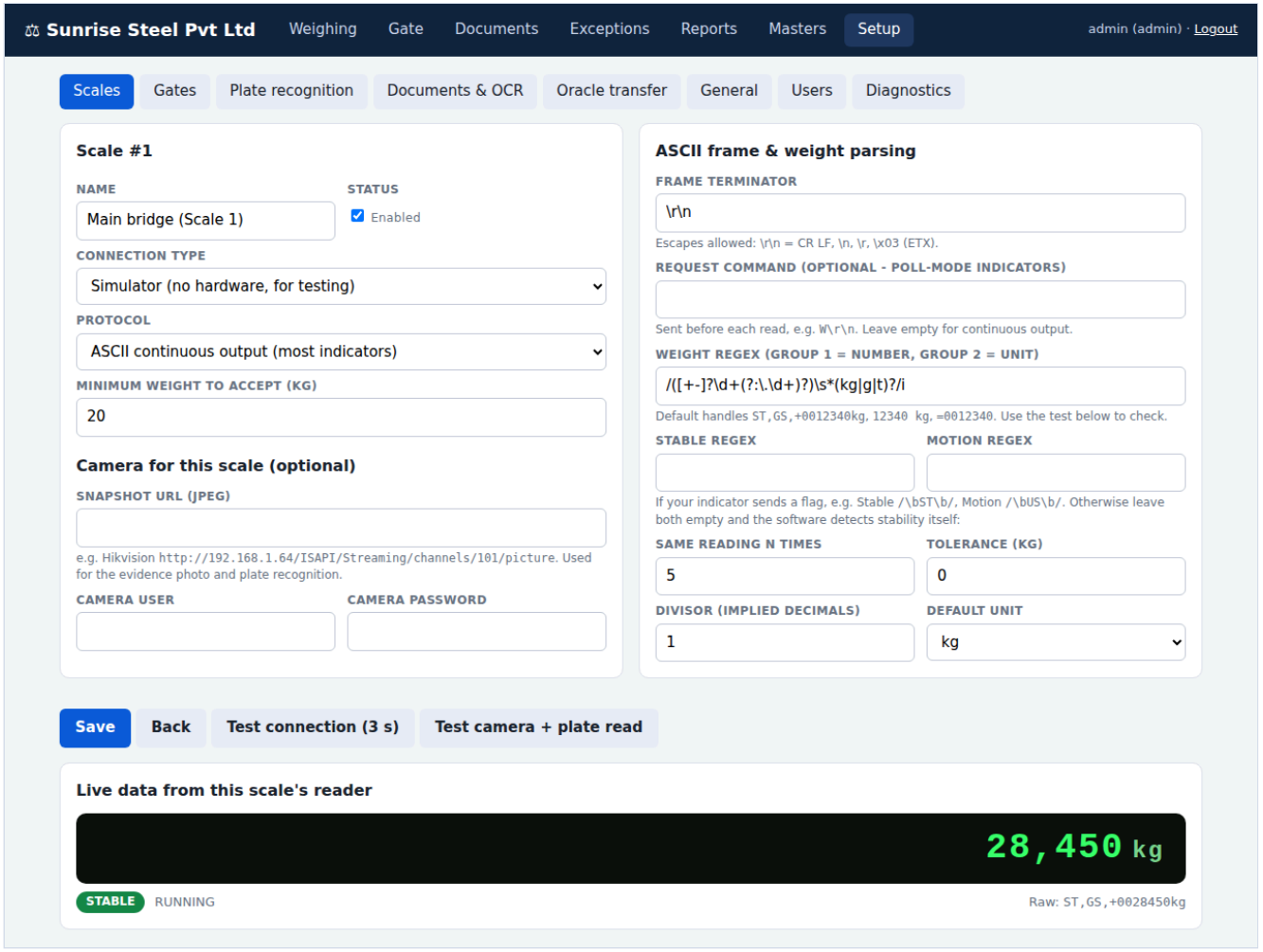
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6. Connecting the weighing scales (indicators)

Open **Setup → Scales**. Every scale on this PC is listed with its connection and reader status.



- **Add a scale** creates a new one (disabled, in Simulator mode). Press **Edit**, configure it, tick **Enabled**, and **Save**.
- A scale's reader starts within about two seconds of being enabled; changes to a running scale are picked up within five seconds. No restart is needed.
- Two scales cannot use the same COM port or gateway address.



6.1 Connection

Field	Meaning
Connection type	*Serial* (COM / ttyUSB — RS-232 or RS-485 through a USB adapter), *TCP/IP* (serial-to-Ethernet gateway in "TCP server" mode), or *Simulator*.
Serial port	Windows `COM3`; Linux `/dev/ttyUSB0` (prefer `/dev/serial/by-id/...`, which does not change after a reboot).
Baud / data bits / parity / stop bits	Must match the indicator exactly (see its manual).
Protocol	*ASCII continuous output* (most indicators) or *Modbus RTU* (RS-485 transmitters / load-cell amplifiers).
Minimum weight to accept	Weights below this are refused (prevents empty-platform tickets).

6.2 ASCII protocol

The indicator sends a line of text many times per second, for example `ST,GS,+0012340kg` followed by CR LF.

Field	Meaning
Frame terminator	Character(s) that end each line: `\r\n` (CR LF, most common), `\r`, `\n`, or `\x03` (ETX).
Request command	Only for indicators that send a weight *when asked*: the command to send (for example `W\r\n`). Leave empty for continuous output.
Weight regex	A pattern that finds the number (group 1) and unit (group 2) in the line. The default `/([+-]?\d+(?:\.\d+)?)\s*(kg|g|t)?/i` handles most formats.
Stable / Motion regex	If the indicator marks stability in the text (for example `ST` / `US`), enter e.g. `/\bST\b/` and `/\bUS\b/`.

Field	Meaning
Same reading N times, tolerance	If the indicator has no stability flag, the weight is called **stable** when the last *N* readings agree within the tolerance.
Divisor	Use when the indicator omits the decimal point (`012340` meaning 1234.0 → divisor 10).
Default unit	Used when the line has no unit.

6.3 Modbus RTU (RS-485)

Slave ID, function (03 holding / 04 input registers), start address (0-based: register 40001 = address 0), one or two registers (16-bit / 32-bit), word order, signed, poll interval. The divisor field (in the ASCII panel) also applies.

6.4 Test

1. **Save**, then press **Test connection (3 s)**. You will see the raw characters received (`<0D><0A>` means CR LF) and the weight the system extracted.
2. If "Could not parse a weight" appears, adjust the terminator or regex until the weight is right.
3. The test needs exclusive use of the port. If the scale is already running, the page tells you: either watch the **Live data** panel at the bottom of the page (it shows what the running reader receives), or disable the scale, wait ten seconds, and test.
4. Put a **known weight** on the bridge and confirm the number on the Weighing screen.

6.5 Wiring notes

- **RS-232:** indicator TX → PC RX, indicator RX → PC TX, GND ↔ GND. If nothing arrives, swap TX/RX.
- **RS-485:** A→A (D+), B→B (D−). Labels differ between vendors; if there is no data, swap the two wires. Add a 120 Ω resistor at the far end of long cables.
- **Indicator setting:** make sure its serial output is set to *continuous* (or the mode you configured), at the same baud/parity.

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7. Cameras and plate recognition (ANPR)

Plate recognition is **optional**. Without it, the operator types the vehicle number.

7.1 Cameras

For every scale (*Setup → Scales → Edit → Camera*) and every gate (*Setup → Gates → Edit → Camera*) you can enter a **snapshot URL** — a web address that returns the current picture as a JPEG — with the camera's user name and password. Typical examples: Hikvision `http://<ip>/ISAPI/Streaming/channels/101/picture`, Dahua `http://<ip>/cgi-bin/snapshot.cgi`. Check your camera manual.

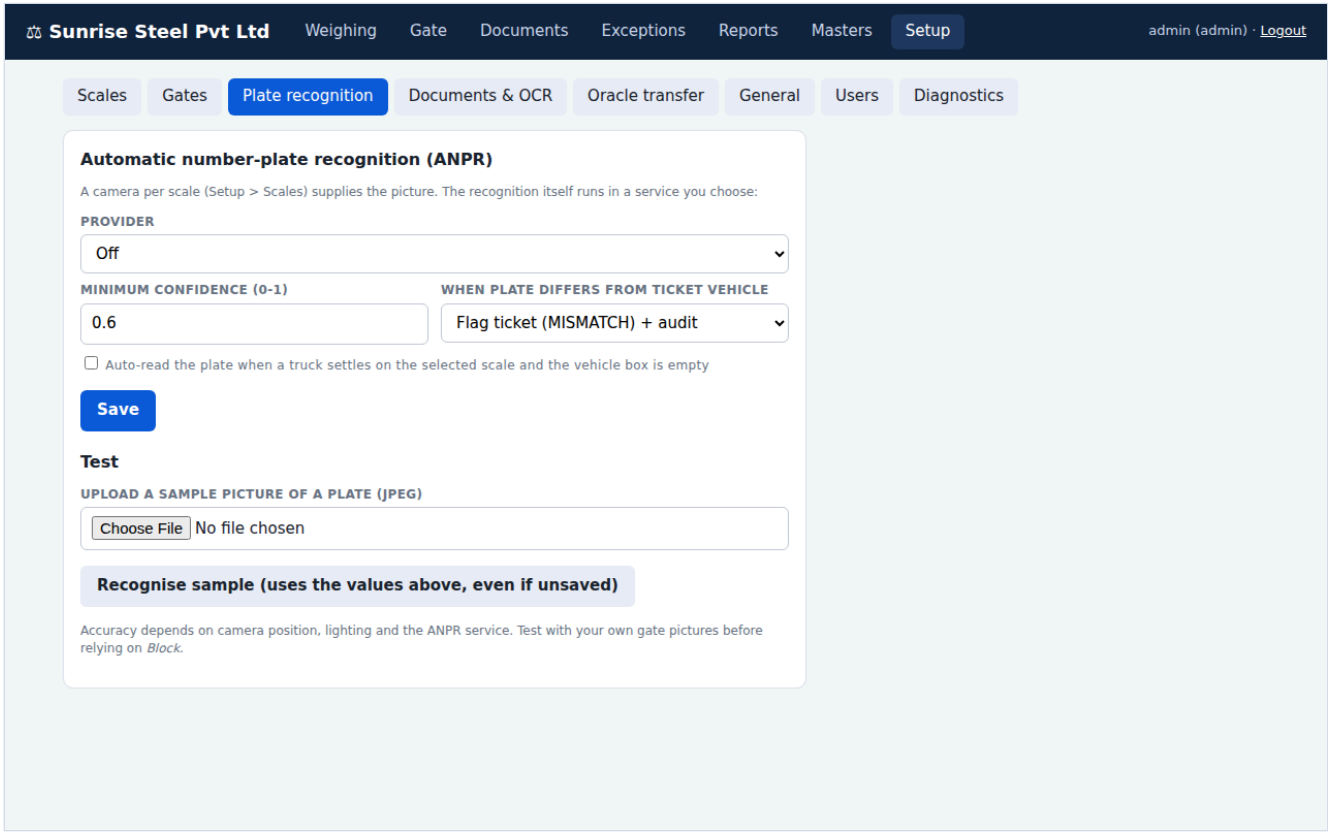
With a camera configured, an **evidence photo** is saved at every first and second weighment and at every gate entry/exit, and shown on the slip / gate pass.

Camera placement (for plate reading): 5–8 m from the plate, angle less than 30°, shutter at least 1/500 s, good lighting or infra-red at night, no headlight glare.

7.2 Recognition service

Setup → Plate recognition. Choose a provider:

Provider	Notes
CodeProject.AI Server	Free, runs on your own PC or server. Install it and its ALPR module; URL `http://127.0.0.1:32168/v1/vision/alpr`.
Plate Recognizer	Commercial. Cloud service (needs internet and an API token) or their on-premises container.
Local command	Any program that prints the plate as JSON or as `PLATE` / `PLATE 0.93`, for example OpenALPR: `alpr -c in -n 1 -j {image}`. Run without a shell for safety.



- **Minimum confidence** (default 0.6): a plate read below this is treated as *unread* rather than risk a wrong number.
- **When the plate differs from the ticket vehicle:** *Ignore*, *Flag ticket* (recorded as MISMATCH, audit entry), or *Block* (an administrator must tick **Override**).
- **Auto-read:** when a truck settles on the selected scale and the vehicle box is empty, the plate is read automatically.
- **Test:** upload a sample photo of a plate and press *Recognise sample* — it works with unsaved values. On a scale's page, **Test camera + plate read** checks the real camera.

7.3 How it is used

- **Read** button next to the vehicle number: fills the number, tells you whether the vehicle is known, its stored tare, and whether it already has an open ticket (or is on the block list, at the gate).
- On the **second weighment** the system checks that the truck on the bridge is the truck on the ticket.
- Results are stored on the ticket/pass and shown as **OK**, **MISMATCH**, **UNREAD** (plate not readable) or **NOIMG** (no camera picture).
- Plate comparison ignores spaces and dashes and look-alike characters (0/O, 1/I, 5/S, 8/B, 2/Z), and tolerates one wrong character on plates of six or more characters.
- A camera or recognition failure **never stops a weighing** — the ticket is created and flagged.

> **Advice:** test on your own pictures first and start with *Flag*, not *Block*.

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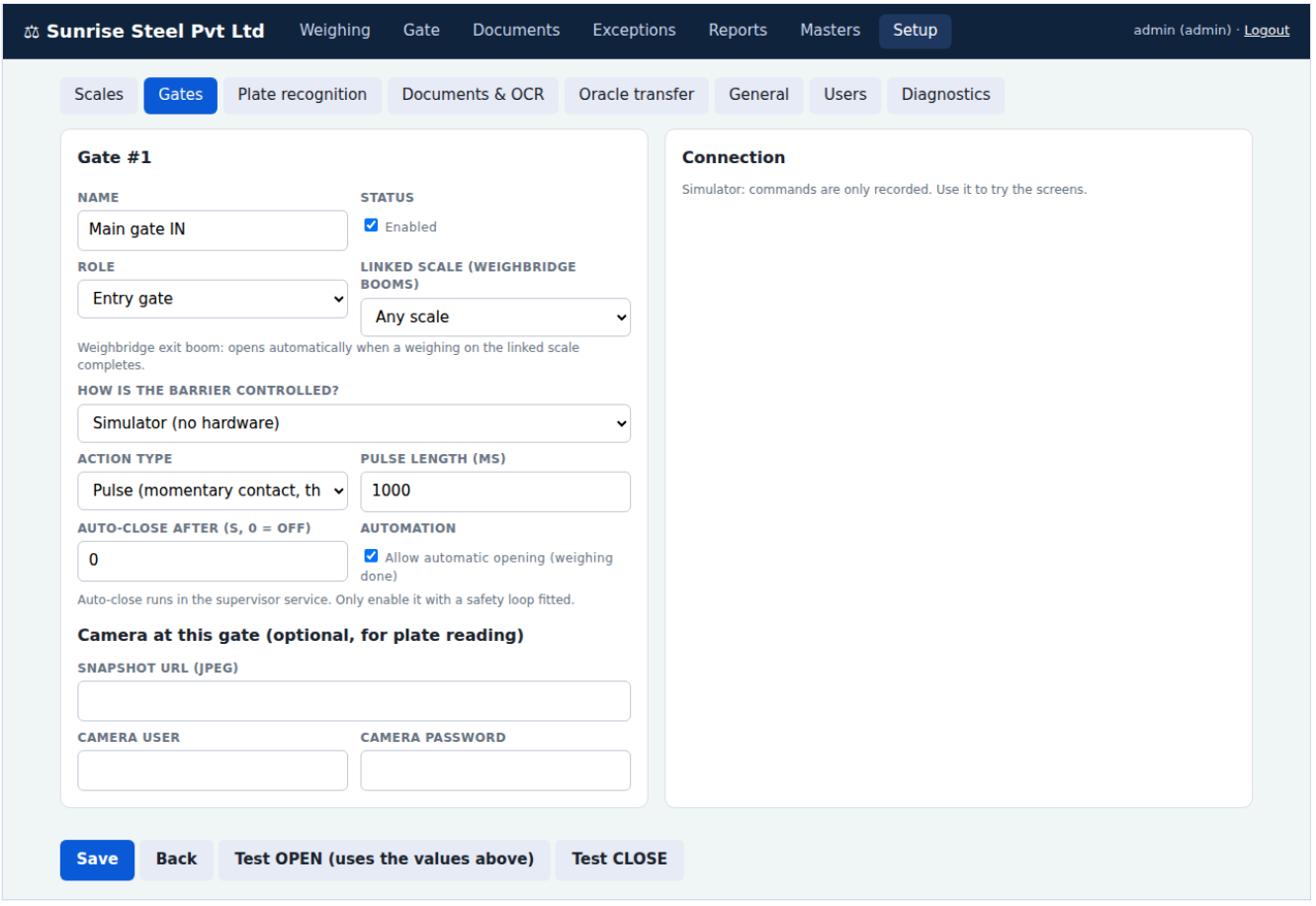
8. Gates and boom barriers

8.1 Safety first

> The program only sends a momentary **contact** (like a remote-control button press) to the barrier controller. It cannot see the barrier and does **not** provide anti-crush protection.
>
> - Fit the barrier's own **safety loop / photocell / limit switches / emergency stop** as required.
> - Connect through a **dry-contact relay** to the controller's OPEN/CLOSE inputs. Never connect PC outputs directly to a motor circuit.
> - **Auto-close** is **off** by default. Use it only when a safety loop is fitted.
> - Test with people and vehicles clear of the barrier.
> - The status badge shows the **last command sent**, not the real barrier position.

8.2 Configure a gate

Setup → Gates → Add a gate → Edit.



Field	Meaning
Role	*Entry gate* / *Exit gate* (used at the security gate), *Weighbridge entry boom*, *Weighbridge exit boom*.
Linked scale	For a weighbridge exit boom: which scale it belongs to (or "Any").
Action type	**Pulse** (momentary contact, then released — most barrier controllers) or **Latched** (relay stays on until the next command).
Pulse length	Milliseconds the contact is held (typical 500–1000).
Auto-close after	Seconds after OPEN before a CLOSE is sent (0 = never). Performed by the scale supervisor service, which must be running.
Allow automatic opening	Lets the system open this gate by itself (weighbridge exit boom opens when a weighing completes).

How the barrier is controlled

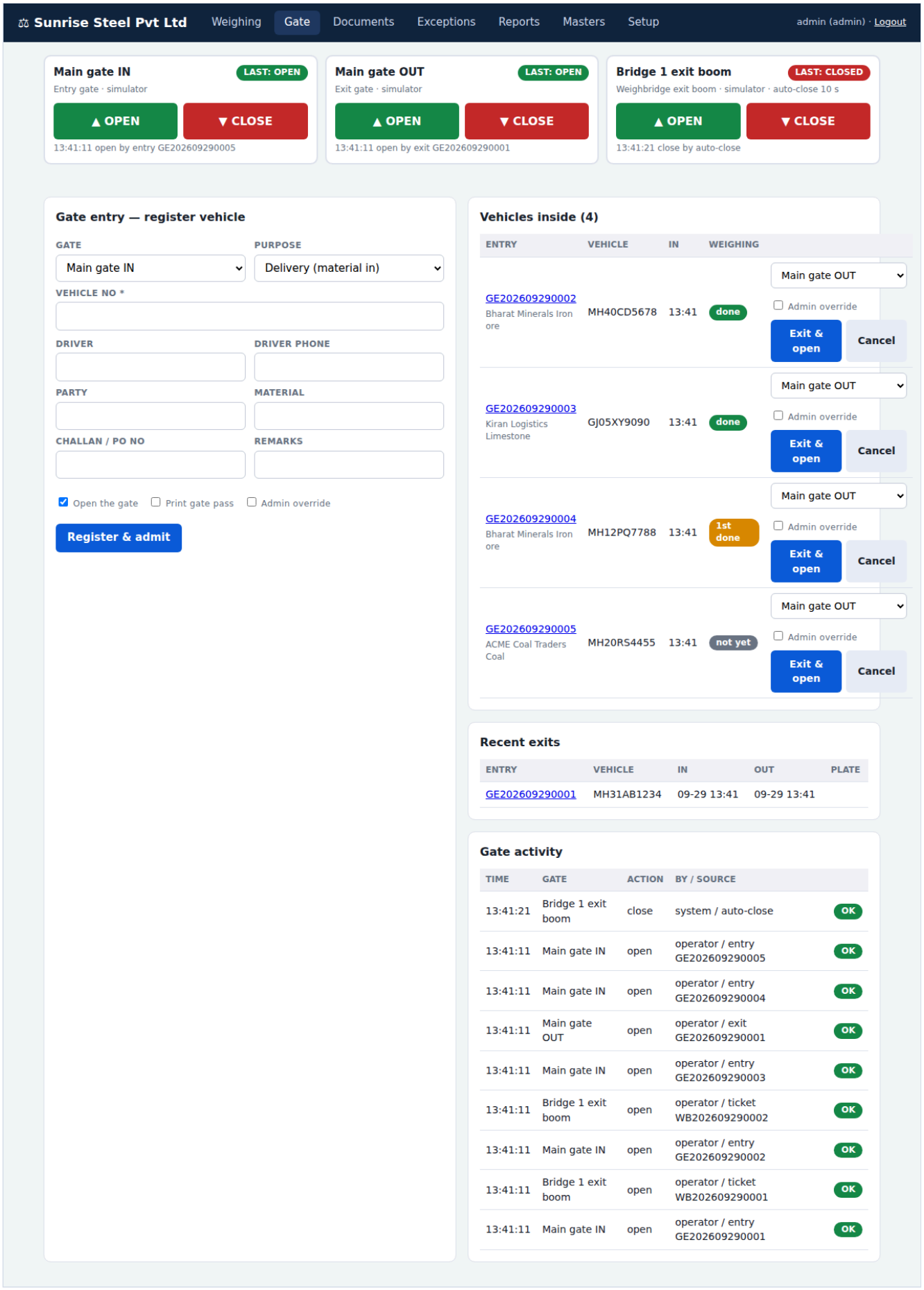
Hardware	Driver	Typical settings
IP relay, Shelly, ESP relay, barrier with web API	**HTTP**	Open URL, Close URL, Release URL (optional), method GET/POST/PUT, optional user/password and body. Success = HTTP 2xx. Example Shelly: open `http://<ip>/relay/0?turn=on`, release `http://<ip>/relay/0?turn=off`.
Ethernet relay board	**TCP**	Host, port, open / close / release commands as text (`OPEN\r\n`) or hex (`hex:A0 01 01 A2`).
USB / RS-232 relay board	**Serial**	Its own COM port, baud (usually 9600); commands as hex, e.g. open `hex:A0 01 01 A2`, release `hex:A0 01 00 A1`.
RS-485 relay module / PLC	**Modbus RTU**	Slave ID, open coil, optional close coil. Needs its own adapter.
PLC / Ethernet I/O module	**Modbus TCP**	Host, port 502, unit ID, coils.
—	**Simulator**	Commands are only recorded; for training.

Modbus coils: *one coil, latched* = ON opens, OFF closes. *Two coils, pulse* = each coil is pulsed.

A gate relay cannot share a COM port with a scale. Two gates may share one relay board (different channels).

Use **Test OPEN** / **Test CLOSE** in the editor (it works before you save) and watch the barrier.

8.3 The Gate screen



1. **Gate cards** — big **OPEN** and **CLOSE** buttons. Every press is logged (who, when, result). If the relay does not answer, the card shows the failure and the state does not change.
2. **Gate entry — register vehicle** (section 11).
3. **Vehicles inside** and **Recent exits**.
4. **Gate activity** — the last commands, including automatic ones (`auto-close`, `ticket WB…`, `entry GE…`).

8.4 Automatic actions

Event	Automatic action
Vehicle registered at an entry gate (with "Open the gate" ticked)	Entry gate opens
Vehicle exits (with "Exit & open")	Exit gate opens
Second weighment (or single-pass ticket) completes on a scale	Weighbridge exit boom linked to that scale opens (if "Allow automatic opening" is ticked)
Auto-close time passes	Gate closes

A barrier fault **never** blocks or undoes a weighing — the failure is shown in Gate activity.

8.5 Gate rules

Setup → Gates → Gate entry rules:

- **Weighing requires a gate entry** — *No* / *Warn* / *Yes – block* (an administrator can override).
- **Exit requires completed weighment and no open ticket** — *No* / *Warn* / *Yes – block*.

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9. Oracle database integration

9.1 What is sent

Every **completed** (closed) ticket, and every **cancelled** ticket that was already sent, is transferred to one Oracle table (default `WEIGHBRIDGE_TICKETS`). The transfer uses an Oracle `MERGE` on the ticket number, so a ticket that is sent twice updates the same row — **no duplicates**. If Oracle or the network is down, tickets simply wait (status *PENDING*) and are sent automatically when it returns. Weighing is never interrupted.

9.2 Prepare Oracle

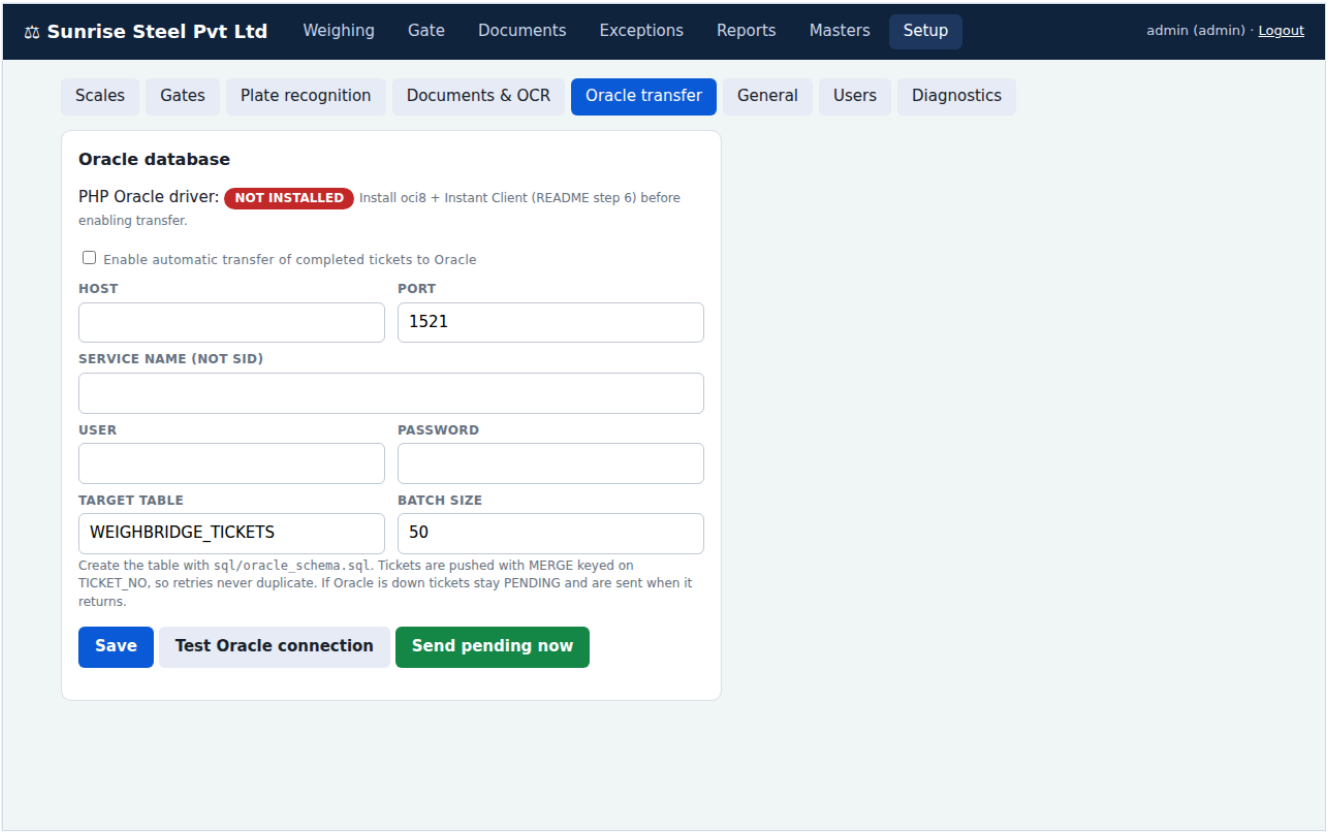
1. Run `sql/oracle_schema.sql` in the schema that will own the table (creates `WEIGHBRIDGE_TICKETS`, primary key `TICKET_NO`, two indexes).
2. Create a dedicated user with only `CREATE SESSION` and `SELECT, INSERT, UPDATE` on that table.
3. If you created the table with an earlier script, run the `ALTER TABLE` statement at the bottom of the file to add `SCALE_NAME`, `PLATE_IN`, `PLATE_OUT`, `PLATE_FLAG`.
4. Make sure the weighbridge PC can reach the listener (default port 1521).

9.3 Prepare the PC

Install **Oracle Instant Client (Basic)** matching PHP's 32/64 bit, and the PHP **oci8** extension (README step 6 has the exact Windows and Linux commands). Check with `php -m` that `oci8` is listed, and run `php bin/check.php`.

9.4 Configure

Setup → Oracle transfer: host, port, **service name** (not SID), user, password (stored encrypted), table name, batch size. Press **Test Oracle connection** (it also checks that the table exists), tick **Enable automatic transfer**, and **Save**. The *WeighbridgeOracleSync* service (or `bin/sync_oracle.php --loop=30`) performs the transfer.



9.5 Monitoring

- *Reports* shows the Oracle status of each ticket: **PENDING** (waiting), **SYNCED**, **FAILED** (hover for the Oracle error), **NA** (Oracle not enabled).
- Administrators can press **Send to Oracle now** or **Re-queue failed**.

- *Setup → Diagnostics* shows the number waiting and the last error.
- `http://<server>/api/health.php` returns `503` when a scale reader is not running — use it with your monitoring system. It also reports how many documents wait to be verified or sent and how many HIGH exceptions are open.

9.6 Columns

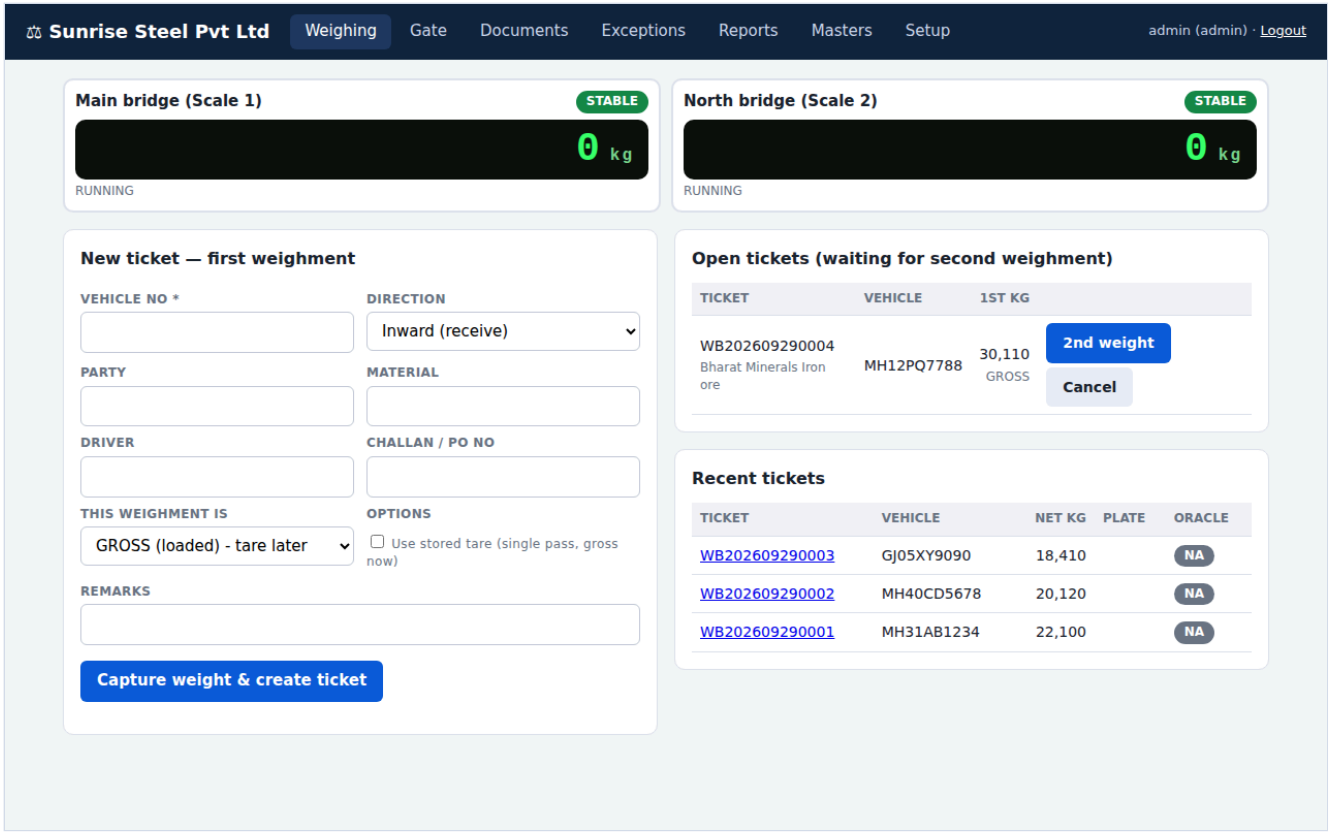
See Appendix B.

9.7 Invoice documents

Verified invoice and return documents can also be written to two more Oracle tables, `WB_DOCUMENTS` and `WB_DOCUMENT_LINES` (script in `sql/oracle_schema.sql`, columns in Appendix B2). The same connection is used; choose the target in *Setup → Documents & OCR* (section 13).

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10. Daily operation — weighbridge operator



10.1 The screen

- At the top, one **live display per scale**: weight in kg and a badge **STABLE** (green), **MOTION** (amber) or **NO DATA** (red). Click the display of the bridge you are standing at; the choice is remembered on that PC.
- The **capture buttons are disabled until the weight is stable**. The weight recorded is the one the system holds at the moment you press the button — it cannot be typed or altered in the browser.
- Left: **New ticket**. Right: **Open tickets** (waiting for the second weighment) and **Recent tickets**.

10.2 Normal two-pass weighing (inward)

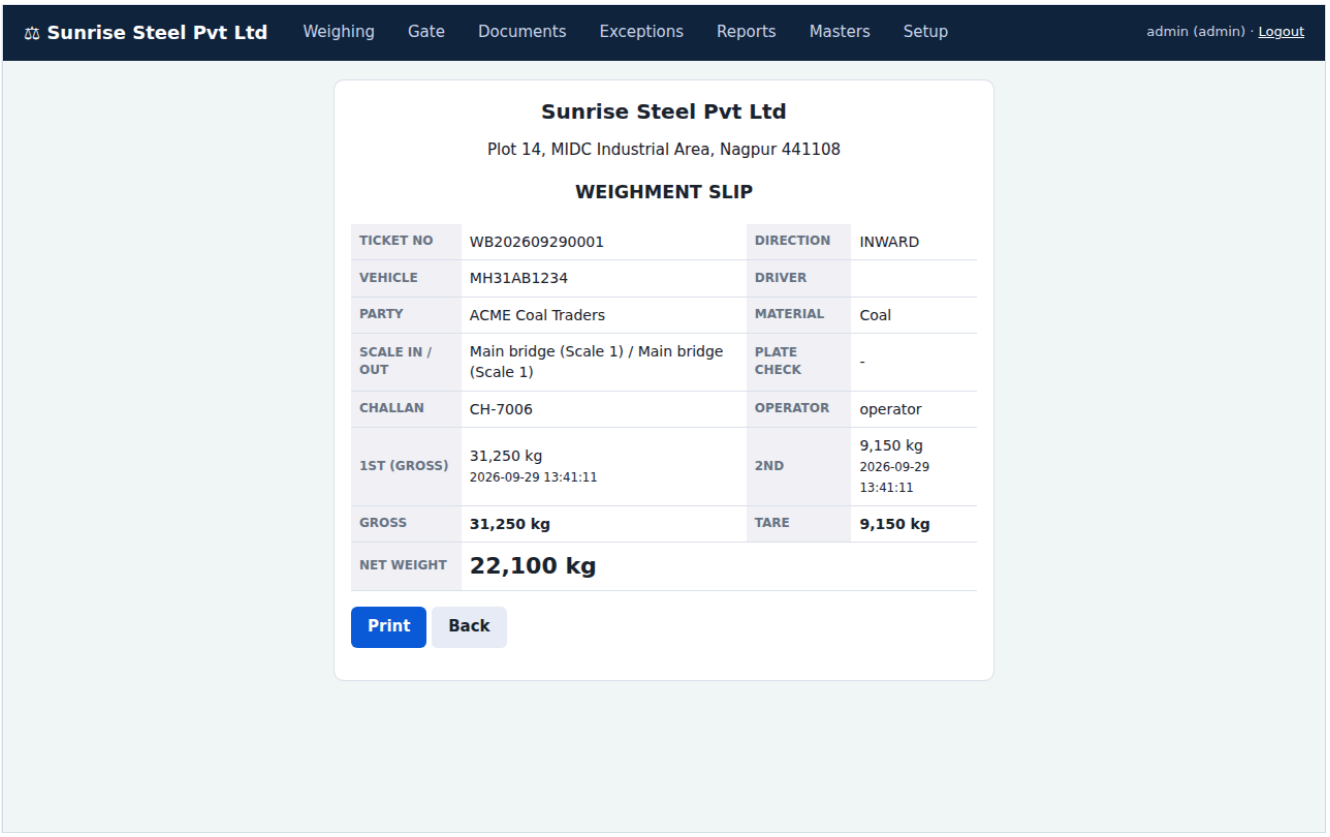
1. Truck drives onto the bridge. Wait for **STABLE**.
2. Enter the **vehicle number** (or press **Read** to read the plate). Choose party and material; driver and challan are optional.
3. Leave *This weighment is* = **GROSS (loaded) – tare later** for a loaded truck arriving, or choose **TARE (empty)** for an empty truck arriving to load.
4. Press **Capture weight & create ticket**. The ticket appears under *Open tickets*.
5. Truck goes away, unloads or loads, and returns.
6. On the returning truck's row press **2nd weight** (with the truck stable on a scale — it can be a different bridge; select it at the top).
7. The slip opens. The system computes **Net = Gross − Tare**, shows all values, and prints on **Print**. The weighbridge exit boom opens if configured.

The system refuses a second weighment if Gross is not greater than Tare (wrong first-weight type) — cancel the ticket (administrator) and redo.

10.3 Single-pass weighing (stored tare)

For vehicles whose empty weight is stored in *Masters*: tick **Use stored tare (single pass, gross now)**, capture the loaded weight; the ticket is completed immediately. The stored tare is learned automatically from the first two-pass weighing of a vehicle.

10.4 Slip



10.5 Messages you may see

Message	Meaning / action
"Weight is not stable yet"	Wait for the STABLE badge.
"No live weight … reader stopped or scale disconnected"	Check the indicator and cable; ask the administrator to look at *Setup → Scales*.
"Weight … is below the minimum capture weight"	Platform empty or truck partly off the bridge.
"Vehicle … already has an open ticket"	Use **2nd weight** on the open ticket instead of creating a new one.
"Gross must be greater than tare"	The first-weight type was wrong; administrator cancels and redoes.
"Camera read plate X but the ticket vehicle is Y"	Plate check (block mode): confirm the vehicle number; only an administrator can override.
"Vehicle … has no gate entry"	Gate rule: register the vehicle at the gate first (or administrator override).

10.6 If the scale is down

If the administrator has allowed **manual weight entry**, a *Manual weight* box appears; enter the weight from another source. Such weights are marked *(manual)* on the slip and should be exceptional.

10.7 Cancelling

Only an administrator can **Cancel** an open ticket (a reason is required and is recorded).

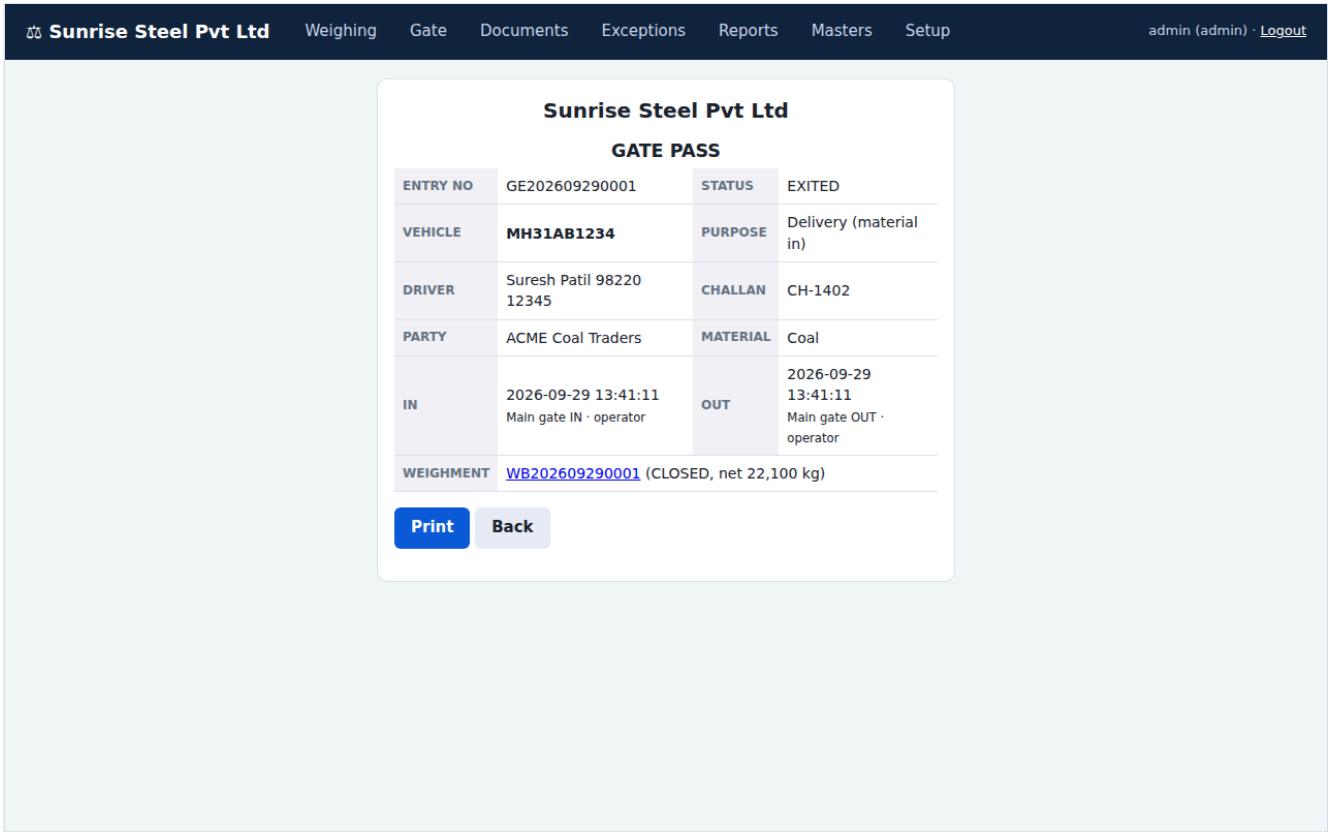
---

11. Daily operation — security gate

Open the **Gate** screen.

11.1 Admitting a vehicle

1. Choose the **entry gate** and **purpose** (delivery, dispatch, visitor/other, empty/returning).
2. Enter the **vehicle number**, or press **Read** to read it from the gate camera. The note under the field tells you if the vehicle is *known*, *new*, **blocked**, or **already inside**.
3. Fill in driver, phone, party, material, challan.
4. Leave **Open the gate** ticked; tick **Print gate pass** if you want a printout.
5. Press **Register & admit**.



Refusals: a vehicle that is already inside, or on the **block list**, is refused and the gate stays closed (the attempt is recorded). An administrator can admit a blocked vehicle by ticking *Admin override*; this is noted on the pass.

11.2 Vehicles inside

The list shows every vehicle currently inside with its **weighing status**: *not yet*, *1st done* (loaded/empty weight taken, second pending) or *done*.

11.3 Letting a vehicle out

1. Find the vehicle under *Vehicles inside*.
2. Check the exit gate in the drop-down and press **Exit & open**.
3. If the exit rule is set to *block*, the exit is refused while the vehicle still has an **open ticket** or (for delivery/dispatch) **no completed weighment**. The message says which. An administrator may override; it is recorded on the pass.

11.4 Opening or closing a barrier manually

Press **OPEN** or **CLOSE** on the gate's card. The result appears under the buttons and in *Gate activity*. Use manual buttons for staff, emergency vehicles and testing.

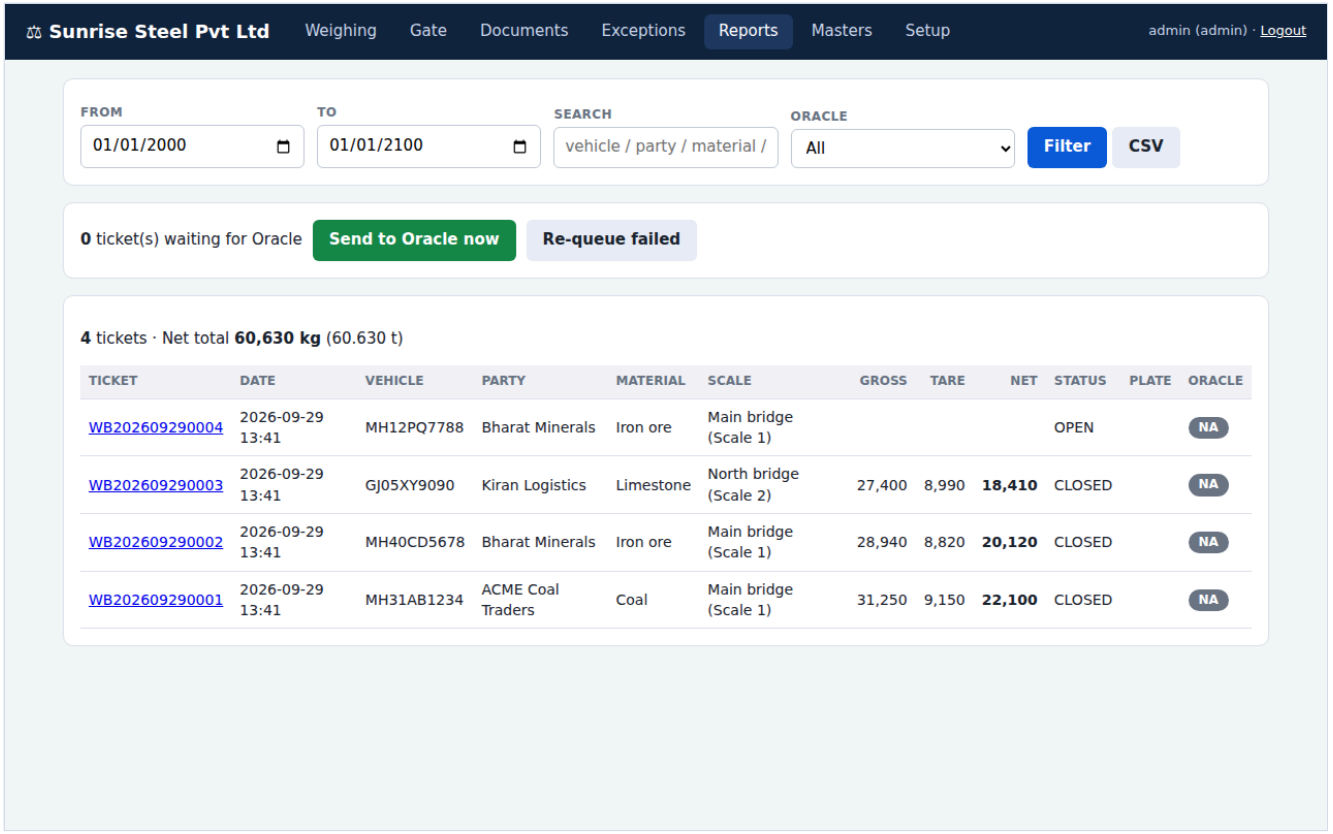
11.5 Blocking a vehicle

Masters → Vehicles → Block (administrator): enter a reason. The vehicle is refused at the gate until unblocked.

---

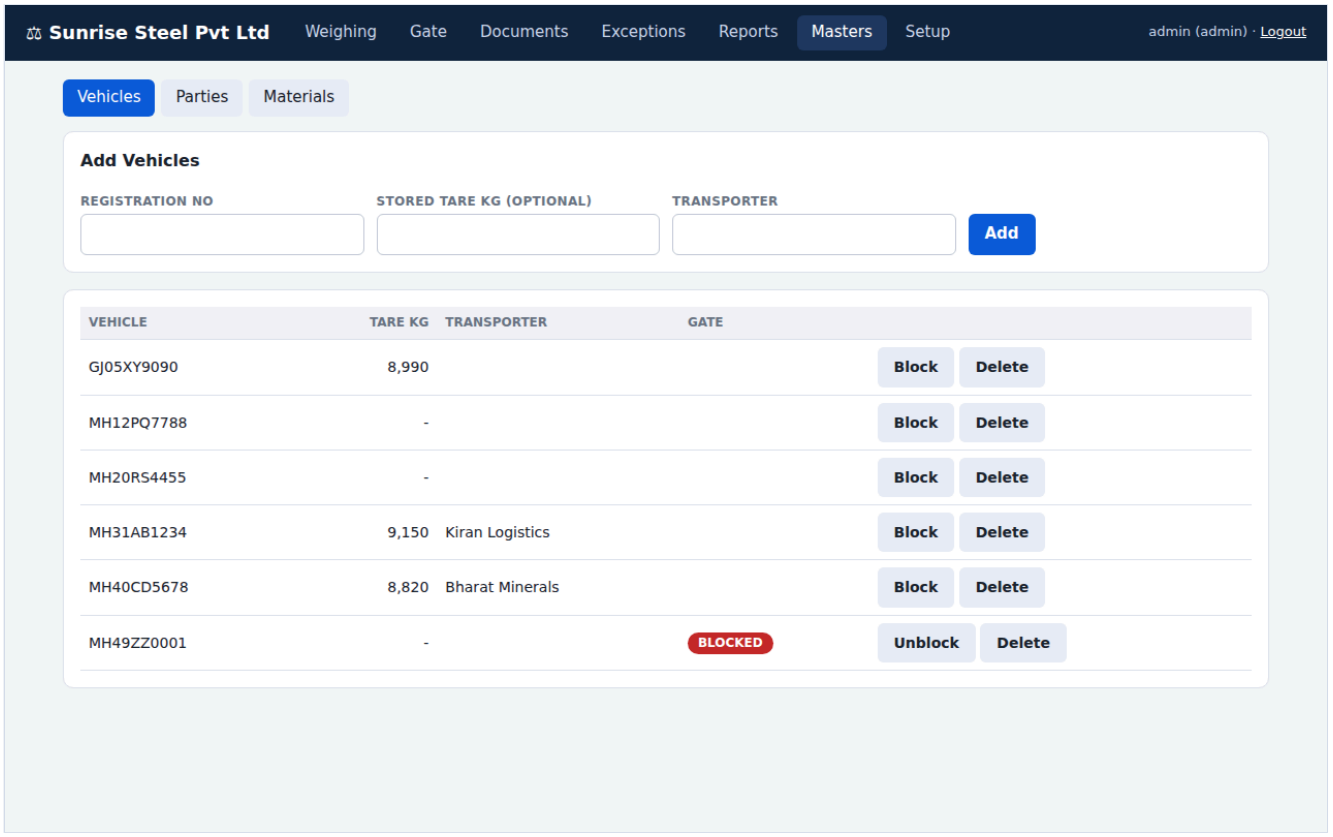
12. Reports, tickets and the ERP interface

12.1 Reports



Reports lists tickets for a date range with search (vehicle, party, material, ticket number) and an **Oracle** filter. It shows totals in kg and tonnes for completed tickets, plate check result, scale, and Oracle status. **CSV** downloads the filtered list for Excel. Click a ticket number to view or reprint the slip.

12.2 Masters



Vehicles (registration, **stored tare**, transporter, blocked flag), parties and materials. Add anyone; administrators can delete and block.

12.3 ERP interface (REST API)

Setup → Odoo & API → REST API → Generate new token (the token is shown **once**; store it safely; it is kept only as a hash). Your ERP then calls:

```
GET http://<server>/api/v1/tickets.php?since_id=0&limit=100
Authorization: Bearer <token>
```

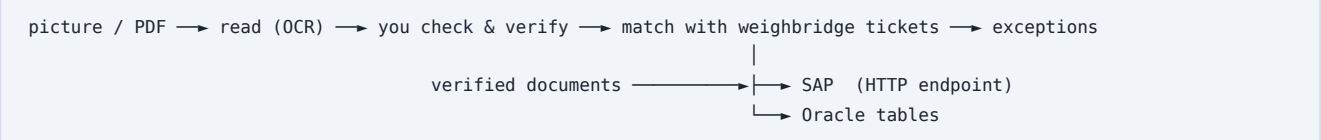
(The token is now generated under *Setup → Odoo & API*.) It returns completed and cancelled tickets with `id` greater than `since_id` (maximum 500 per call) as JSON. The ERP remembers the highest `id` received and asks for newer ones next time. *Disable API* revokes access. Documents, purchase orders and master data are also available – and, if you allow it, writable – see **chapter 17**, which also describes the direct connection to **Odoo**.

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13. Invoice and return documents (OCR, matching, exceptions)

13.1 What it does

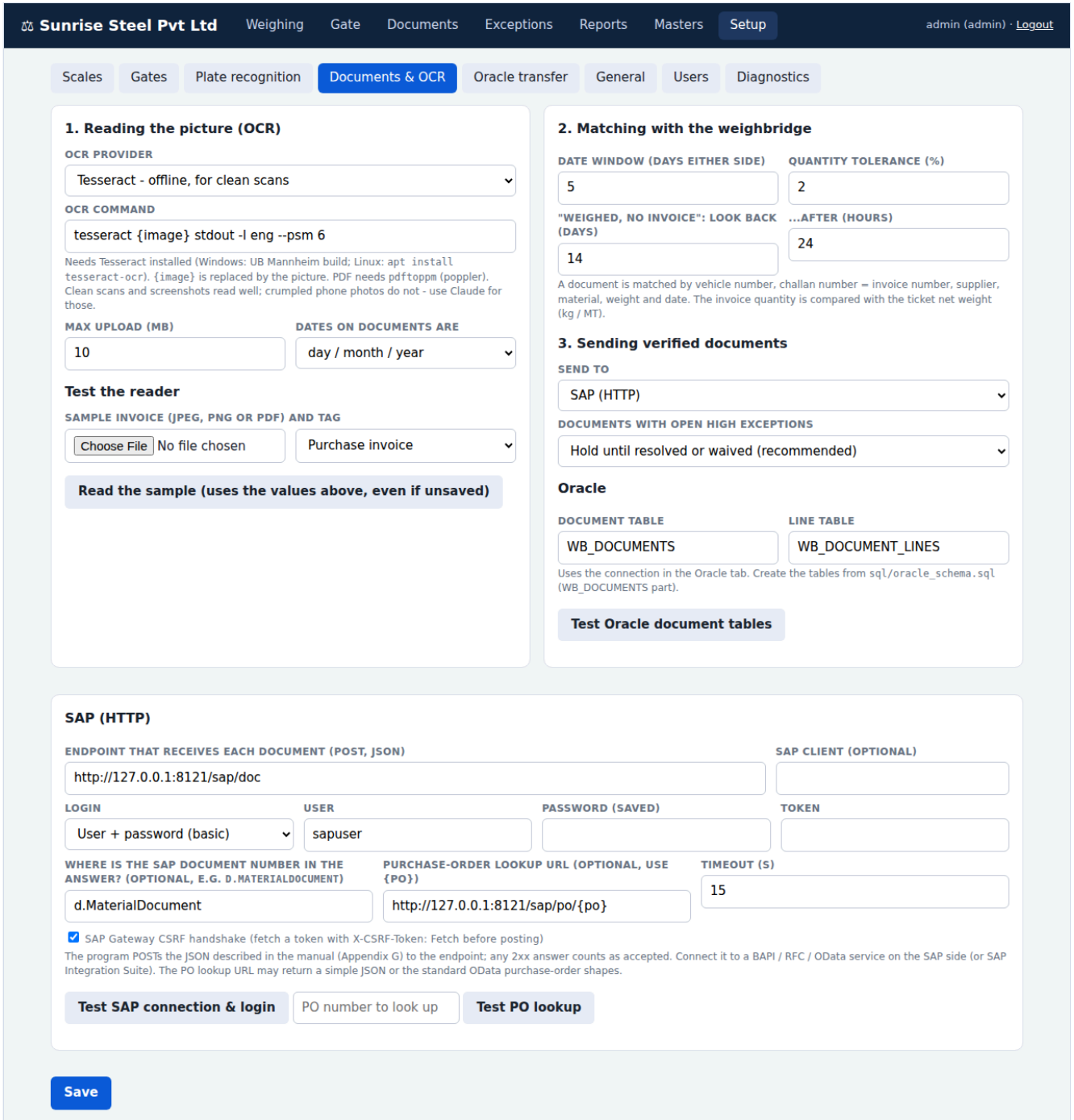
Purchase invoices and material-return documents arrive on paper or as pictures. This module turns each one into a structured **document**, checks it against what the weighbridge actually recorded, and passes it on to SAP and/or Oracle:



- **Two tags**: *Purchase invoice (PO)* – goods received against a purchase order; *Material return* – goods sent back to the supplier (return challan, debit note, rejection note).
- **A person always verifies.** OCR can misread a digit. The reader's result is shown next to the picture, unsure fields are highlighted, and nothing counts as "verified" until someone presses **Save & Verify**.
- **The weighbridge is the referee.** The quantity on the invoice is compared with the net weight of the linked ticket(s); the truck, supplier, material and date are compared too.

13.2 Setting it up (administrator)

Setup → Documents & OCR.



1. Choose how pictures are read

	Claude (Anthropic API)	Tesseract (offline)
Good at	Phone photos, skewed or creased pages, PDFs, any layout, units and line items	Clean scans and screenshots
Weaker at	Needs internet	Photos and unusual layouts; often loses the unit (MT/KG) or cannot find line items
Where the picture goes	**Sent to the Anthropic API**	Stays on your PC
Cost	Pay per use to Anthropic	Free
Set-up	API key (and optionally a model name)	Install Tesseract; nothing else

Try both with about ten of your own invoices before deciding: use **Read the sample** on the same page (it works before saving). Setting *Thinking effort* to *low* makes Claude faster and cheaper; *high* helps very messy documents. The model name can be

changed if you prefer a different one. On Windows install the "UB Mannheim" build of Tesseract; `php bin\check.php` tells you whether it is found. PDF files with Tesseract also need `pdftoppm` (poppler) – otherwise upload JPEG/PNG or use the Claude reader.

2. Matching rules – date window (default ±5 days), quantity tolerance (default 2 %), and how long a weighed truck may go without an invoice before it is listed (default 24 hours, looking back 14 days). Dates on your documents are read day-first unless you change it.

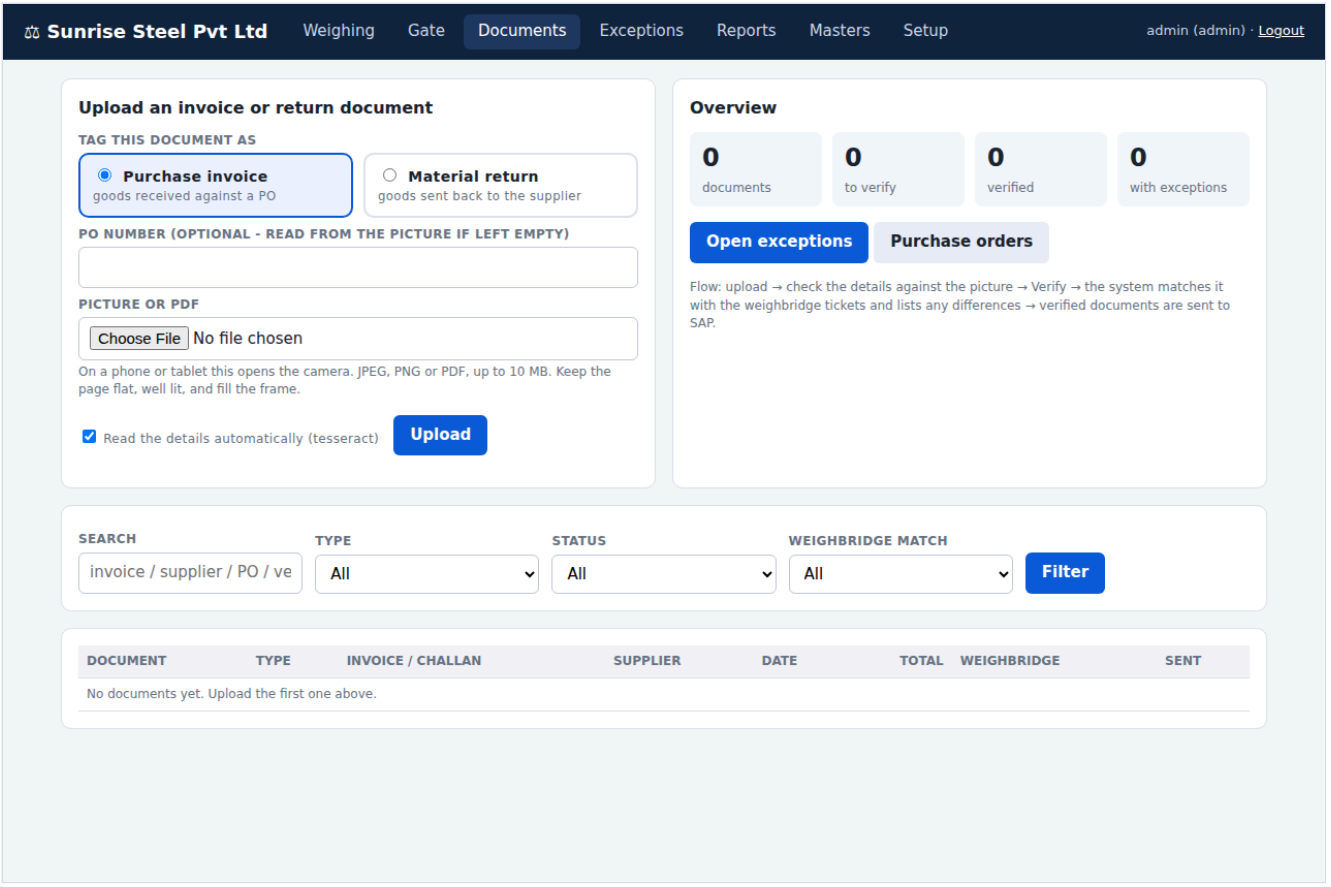
3. Where verified documents go – *Nowhere*, *Oracle tables*, *SAP (HTTP)* or both; and whether documents with open HIGH exceptions are **held** (recommended) or sent anyway.

4. SAP – see 13.9. **5. Oracle** – table names (see 9.7).

6. PHP upload limits – make sure `upload_max_filesize` is at least 10M and `post_max_size` at least 12M in `php.ini` (`php bin/check.php` warns if not).

13.3 Uploading a document

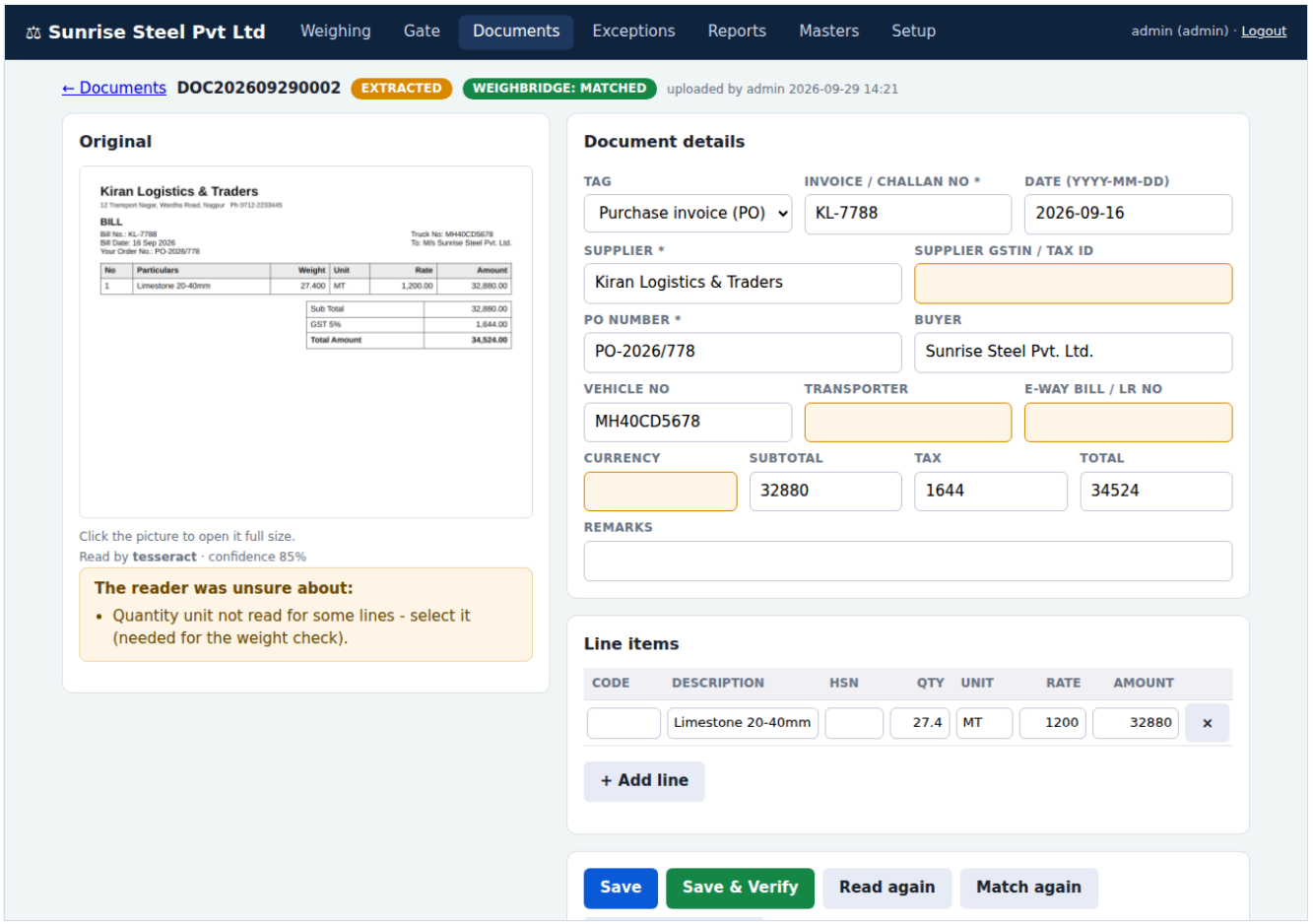
Documents → **Upload an invoice or return document**.



1. Choose the tag: **Purchase invoice** or **Material return**.
2. Optionally type the PO number (or, for a return, the original invoice / return reference). If left empty it is read from the picture.
3. Choose the file. **On a phone or tablet this opens the camera.** JPEG, PNG or PDF, up to 10 MB (changeable).
4. Leave *Read the details automatically* ticked and press **Upload**. Reading takes a few seconds (Tesseract) up to about a minute (Claude with long PDFs). You are taken to the review screen.

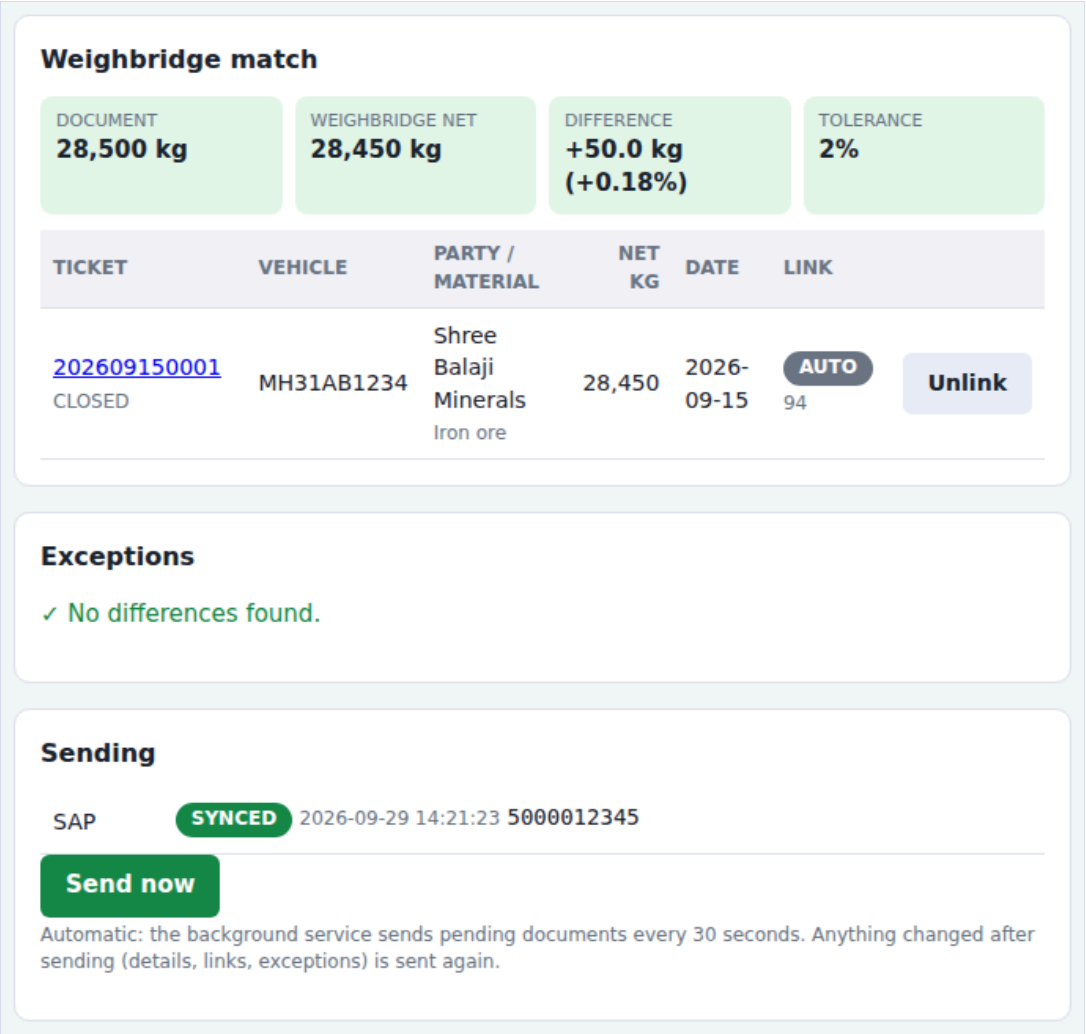
Photo tips: put the page flat on a plain surface, fill the frame, use good light without shadow or glare, hold the phone parallel to the page, and make sure the invoice number, date, quantity and totals are sharp. One file = one document; a multi-page invoice should be uploaded as one PDF.

13.4 Reviewing and verifying



- **Left:** the original picture (click to open it full size) and the reader's own warnings ("The reader was unsure about…").
- **Right:** the details in editable fields. **Yellow fields are empty** (or a quantity has no unit). Compare every important number with the picture: invoice number, date, supplier, PO number, vehicle number, each line's quantity, **unit**, rate and amount, and the totals.
- **The unit matters.** The weight check needs a unit such as MT or KG. If OCR could not read it, choose it on each line. (The exceptions box suggests the likely unit from the weighbridge weight: "The weighbridge weight suggests it is MT".)
- Wrong tag? Change *Tag* – the PO field or the original-invoice fields appear accordingly.
- Add or remove line items with **+ Add line** and **×**.

Buttons: **Save** (keep working on it), **Save & Verify** (locks the document, matches it, queues it for sending), **Read again** (replaces the fields with a fresh reading), **Match again**, and for administrators **Cancel document**. A verified document can be **Reopened** for corrections; it is sent again if anything changed.



13.5 Matching with the weighbridge

Every time a document is saved, verified or re-matched, the system looks for the weighbridge ticket(s) it belongs to. Purchase invoices are matched to **INWARD** tickets, material returns to **OUTWARD** tickets.

Evidence	Points
Challan number on the ticket equals the invoice number	60
Vehicle number identical (look-alike letters/digits such as 0/O and 1/I count as the same)	50
Vehicle number differs by one character	30 (a possible misread – needs other evidence)
Supplier resembles the ticket's party	up to 15
Invoice item resembles the ticket's material	up to 10
Invoice quantity equals the ticket net weight within the tolerance	15
Ticket date within a day of the document date	5

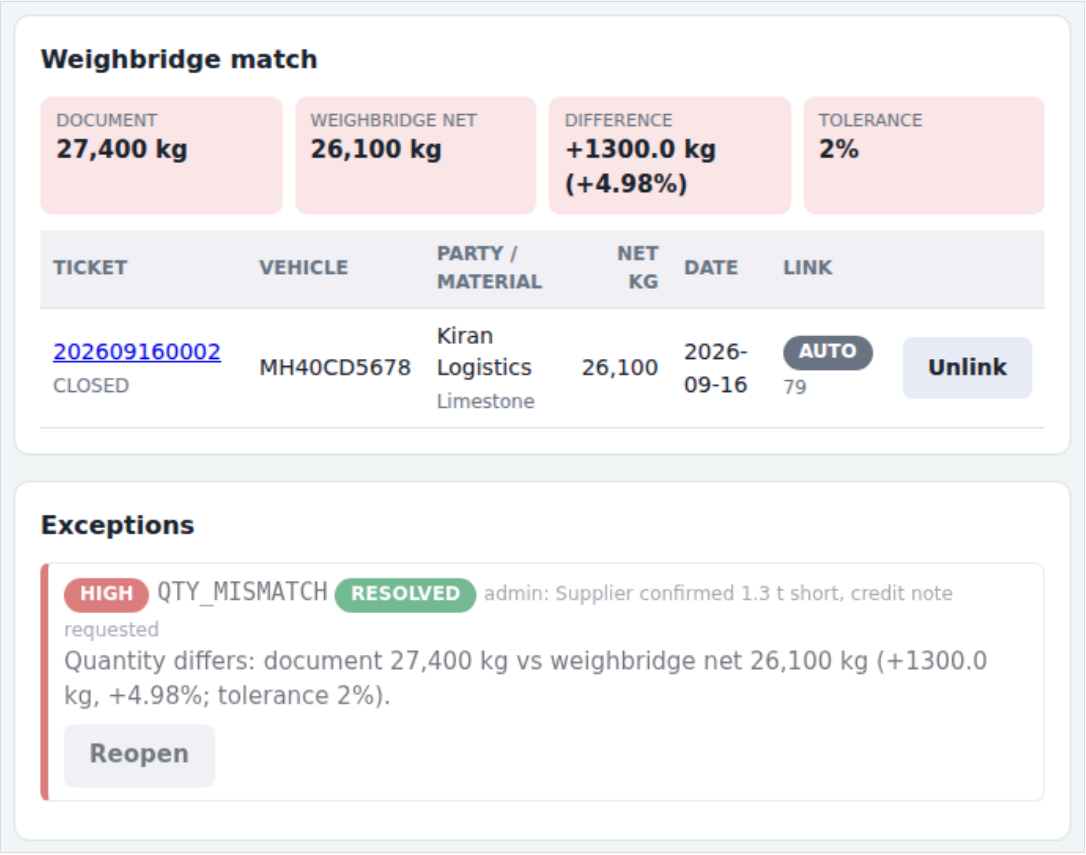
A ticket scoring **50 or more** is linked automatically (marked **AUTO**). If the invoice quantity is larger than the first ticket, further trips of the same truck are added until the quantity is reached (a **split load**). A ticket already linked to another document is not taken over. **Possible tickets** lists other candidates with the reason; press **Link** to link one by hand (**MANUAL**) or **Unlink** to remove a wrong one – the system will not put an unlinked ticket back by itself.

Quantity check. The invoice quantity in kilograms (MT, KG, QTL and grams are converted) is compared with the total net weight of the linked tickets. *Service and charge lines* (loading, handling, freight, HSN codes starting 99…) are not counted as goods, so "Iron ore 28.5 MT" plus "Loading charges 28.5 MT" is compared as 28.5 MT. Items counted in pieces, litres or bags cannot be compared with a weight; they are marked "quantity not checked". A difference within the tolerance is fine; a larger one raises **QTY_MISMATCH** and shows both numbers.

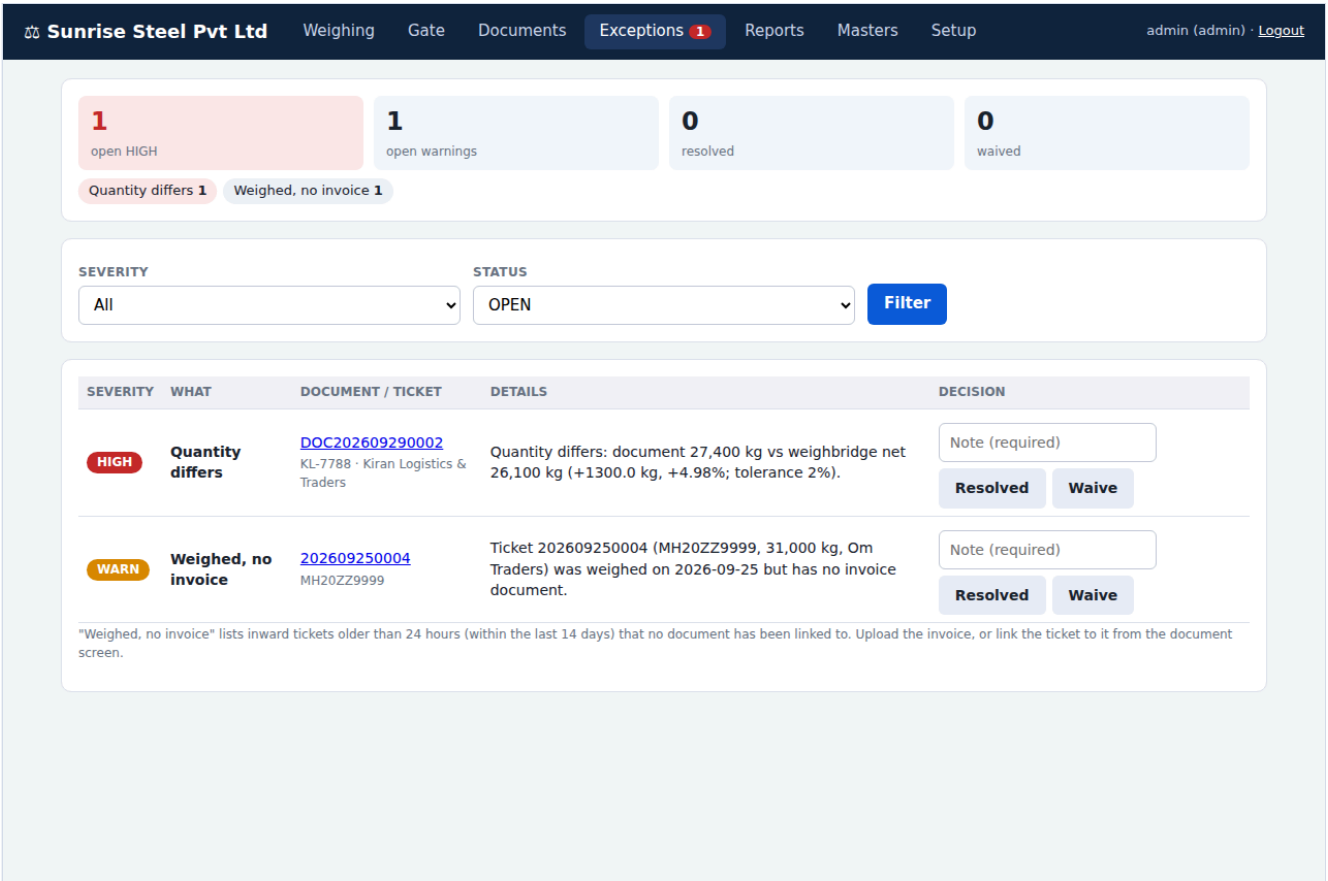
The **match status** shown on the document and in the list:

Status	Meaning
MATCHED	Ticket(s) linked and no open HIGH exception or warning
REVIEW	Ticket(s) linked; only warnings are open
EXCEPTION	At least one open HIGH exception
UNMATCHED	No ticket linked

13.6 Exceptions



Each difference is an **exception** with a severity: **HIGH** (money or goods at risk – holds sending by default), **WARN** (check it) or **INFO** (for your information). The complete list with what to do is in Appendix H. Exceptions appear on the document and on the **Exceptions** screen; the menu shows the number of open HIGH exceptions.



To deal with one, write a short **note** and press:

- **Resolved** – you looked into it and it is dealt with (for example "Supplier confirmed 1.3 t short, credit note requested"). Any user can do this.
- **Waive** *(administrator)* – accept the difference and release the document.
- **Reopen** – undo the decision.

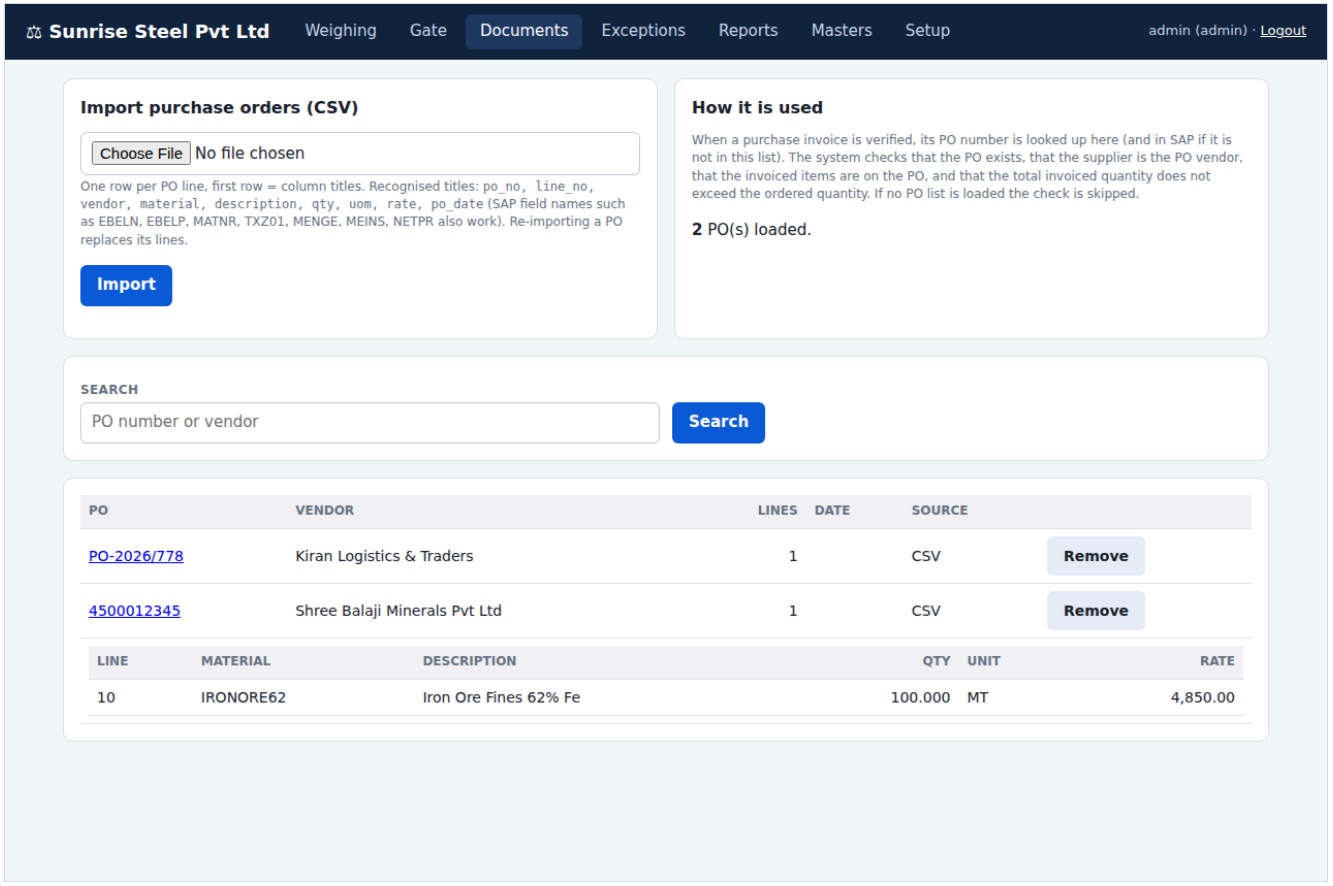
Your decision, name and note are kept and travel with the document to SAP/Oracle. If the underlying cause disappears (a ticket is linked, a quantity corrected) the exception is removed automatically; if the same problem persists, your decision is kept and it is not raised again.

Weighed, but no invoice. The Exceptions screen also lists **inward tickets that no document has been linked to** after the grace period (default 24 hours, last 14 days). Upload the invoice, or link the ticket to the document, and it drops off the list. This catches deliveries that arrived without paperwork and invoices that never came.

13.7 Purchase orders (PO list)

Purchase invoices are checked against the purchase order. *Documents → Purchase orders* imports a **CSV** export of your open POs (one row per PO line, first row = column titles). Recognised titles: `po_no, line_no, vendor, material, description, qty, uom, rate, po_date` – SAP field names such as `EBELN, EBELP, NAME1, MATNR, TXZ01, MENGE, MEINS, NETPR` are understood too. Comma, semicolon or tab separated. Importing a PO again replaces its lines.

If the PO is not in the list and a **PO lookup URL** is configured for SAP (13.9), it is fetched from SAP on demand and remembered. If neither a list nor a lookup exists, the PO is simply not verified (INFO).



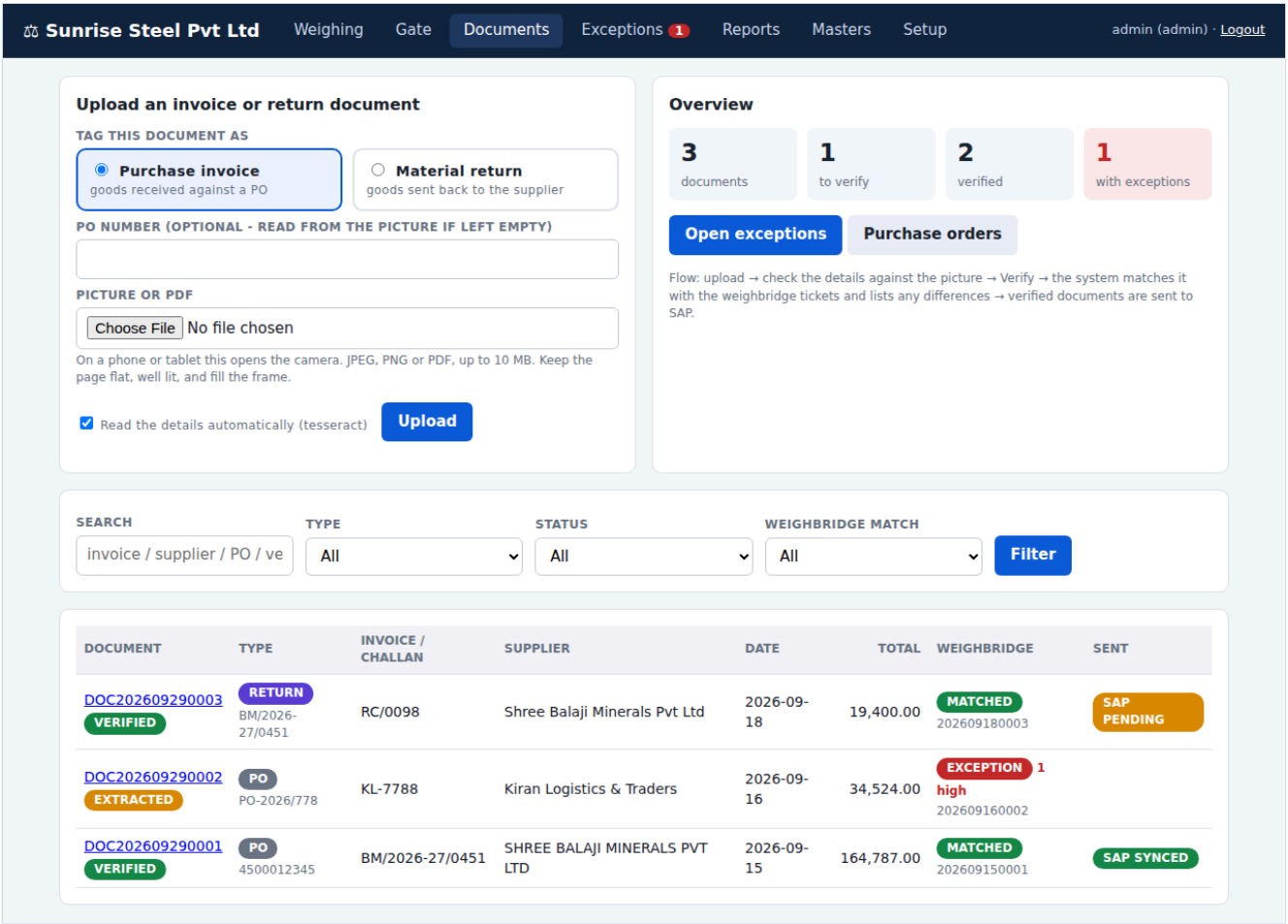
Checks made: the PO number is present; the PO exists; the supplier is the PO's vendor; each invoiced item is on the PO (by material code or by description); the rate matches the PO rate (within 1 %); and the **total quantity invoiced against the PO line by all documents** does not exceed the ordered quantity (plus tolerance).

13.8 Material returns

Tag the document *Material return*. The system reads the **original invoice number** it refers to and checks: the return names an invoice; that invoice is among the uploaded documents; the return is to the same supplier; the returned items are on that invoice; and the **total returned** (this and earlier returns) does not exceed the quantity invoiced. It is matched with an **OUTWARD** weighbridge ticket of the truck that carried the material back.

13.9 Sending to SAP and Oracle

Only **verified** documents are sent. In the document list the columns show **ORA / SAP** status: PENDING, SYNCED, FAILED (hover for the reason) or HELD.



- **Held.** With the recommended policy a document with open HIGH exceptions is **held**: resolve or waive them and it is released. An administrator can tick *send anyway* (the exceptions are included in the data).
- **Automatic.** The background service sends pending documents every 30 seconds and retries failures. **Send now** on the document sends immediately.
- **Changes are re-sent.** If anything that is part of the data changes after sending (edits, linked tickets, an exception decision), the document becomes PENDING and the update is delivered.
- **Independent targets.** If Oracle is down but SAP is up, SAP is still updated; each target has its own status and retry.

SAP. The program **POSTs the document as JSON** (Appendix G) to an HTTPS endpoint you provide, and treats any 2xx answer as accepted. On the SAP side your team connects that endpoint to whatever should happen – for example an OData / ICF service, an SAP Integration Suite / API Management flow, or a BAPI or RFC wrapped in an HTTP service (invoice verification, a staging table, a workflow…). *Setup → Documents & OCR → SAP* takes:

Field	Meaning
Endpoint	The URL that receives each document
Login	User + password (basic), a bearer token, or none. The SAP client number is added as `sap-client` if given
CSRF handshake	Tick for SAP Gateway services: the program first fetches a token (`X-CSRF-Token: Fetch`) and posts with it and the session cookie
Document number path	Where SAP's own document number is in the answer (for example `d.MaterialDocument`); it is stored on the document
PO lookup URL	Optional: a URL with `{po}` that returns the purchase order (a simple JSON, or the standard OData purchase-order shapes)

Use **Test SAP connection & login** (it sends no document) and **Test PO lookup** on the same page.

Oracle. Two tables (`WB_DOCUMENTS`, `WB_DOCUMENT_LINES`) hold the document, its lines, the linked ticket numbers, the weighbridge net weight, the match status and the open exception codes (Appendix B2). Rows are written with an idempotent MERGE.

13.10 Good practice and limits

- **Always verify against the picture.** OCR is a helper, not an authority. Pay most attention to numbers.
- **Start with the hold policy on** and review the exceptions daily. Adjust the tolerance to your real weighing accuracy (2 % is a starting point).
- **Privacy.** With the Claude reader the pictures are sent to the Anthropic API. Use Tesseract if invoices must not leave your premises. Uploaded pictures are stored in `data/documents` (not reachable from the web; shown only to logged-in users).
- **What is not done.** The program does not post accounting entries in SAP itself – it hands the data to the endpoint you provide. It reads one file per document, does not read handwriting reliably, and cannot judge whether an invoice is commercially correct beyond the checks above.
- **Test before go-live.** Read ten real invoices with each reader, check every field, and run one document through SAP / Oracle into a test system (Appendix D, items 21–26). The SAP connection has been exercised against a simulated SAP Gateway (basic login, CSRF token, session cookie, OData purchase-order shapes) but not against a live SAP system – your SAP team should confirm the endpoint on site.

---

14. Administration

14.1 Users

Setup → Users: add users, enable/disable, set passwords. The administrator cannot disable their own login. After 8 failed logins for a user name or address, further attempts are refused for 10 minutes.

14.2 Backups — do this daily

```
php bin/backup.php --dir=D:\Backups\weighbridge --keep=30
```

Creates a consistent copy of the database (safe while the system is running), verifies it, and keeps the newest 30. Schedule it daily: Windows *Task Scheduler*, or Linux cron `15 2 * * * php /var/www/weighbridge/bin/backup.php --dir=/backup/weighbridge --keep=30`.

Also keep a separate copy of **`data/app.key`** — without it the saved Oracle, camera and relay passwords cannot be decrypted (you would re-enter them). Copy the `data/snapshots` folder if you need to keep evidence photos.

14.3 Restoring / moving to another PC

1. Install the program on the new PC (section 4) but do not run the installer page.
2. Stop the services. Copy your backup file to `data/weighbridge.sqlite` (delete any `weighbridge.sqlite-wal/-shm` files) and copy `app.key` to `data/`.
3. Run `php bin/check.php`, start the services, log in.
4. Re-check the COM port names in *Setup → Scales / Gates* (they can differ on a new PC).

14.4 Upgrading to a new version

1. Take a backup (13.2).
2. Stop the services.
3. Copy the new program files over the old ones, **keeping the `data` folder**.
4. Run `php bin/check.php`, then start the services. The database upgrades itself automatically the first time it is opened.
5. Optional: `php tests/run.php` runs the self-tests in a temporary folder (it never touches your data).

14.5 Logs and health

What	Where
Reader logs (one per scale)	`data/logs/scale_<n>.log` (rotated at 5 MB)
Supervisor / service messages	Linux: `journalctl -u weighbridge-scale`; Windows: `data\scale.log`
Audit trail (logins, tickets, cancels, overrides, setup changes)	*Setup → Diagnostics → Recent audit*
Gate commands	*Gate → Gate activity*
System status	*Setup → Diagnostics*, and `api/health.php`

Camera photos in `data/snapshots` are never deleted automatically; remove old ones according to your retention policy.

14.6 Time and date

Tickets use the PC clock and timezone. The default timezone is `Asia/Kolkata`; change it by setting the environment variable `WB_TZ` (for example `WB_TZ=Africa/Nairobi`) for the services.

---

15. Security

- Passwords are stored as one-way hashes. Oracle, camera and relay passwords and the plate-recognition token are stored **encrypted** (key in `data/app.key`).
- Every form is protected against cross-site request forgery; every database query is parameterised; browser security headers are set; repeated failed logins are locked out.
- Weights are captured **on the server** from the reader; the browser cannot supply a weight (except the optional, flagged manual entry).
- Setup, user management, overrides and cancellations are administrator-only and audited.
- Uploaded invoice files are checked (real file type, size, picture dimensions), stored under random names outside the web folder, and shown only to logged-in users. The OCR command and SAP/Oracle addresses can only be set by an administrator; the OCR command is started without a shell.
- Invoice pictures are sent to a third party only if the Claude reader is selected (Setup shows a warning).
- Serial port names and network addresses entered in Setup are validated so they cannot be used to run commands.
- **Recommendations:** keep the PC on a protected network; use HTTPS (reverse proxy) if the screens are reachable beyond the plant network; use individual logins; back up daily; restrict who can open the `data` folder; fit the barrier's hardware safety devices.

---

16. Troubleshooting

First run `php bin/check.php` and look at *Setup → Diagnostics*.

Symptom	Likely cause and fix
Display shows `---`, badge **NO DATA**	Reader not running (start the service / `start.bat`), wrong port or settings, cable. Use *Setup → Scales → Test connection*.
"Serial device not found" / "stty failed"	Wrong port name; on Linux add the service user to `dialout`; unplug and re-plug the adapter.
Test: port opens but no data	TX/RX swapped (RS-232), A/B swapped (RS-485), wrong baud/parity, indicator not in continuous mode (use *Request command*).
Bytes arrive but "could not parse a weight"	Fix terminator (`\r`, `\n`, `\r\n`, `\x03`) and weight regex.
Weight never becomes STABLE	Enter the indicator's stable flag in the Stable regex, or raise tolerance / lower "same reading N times".
Value 10× too large or small	Set **Divisor** (10, 100…).
Modbus: "short/no response"	Slave ID, baud/parity, A/B wiring, address base (0-based).
Modbus: CRC error	Baud/parity mismatch, noise, missing terminator resistor.
Windows: port busy	Another program (the indicator's own utility) holds the COM port.
Scale shows "Reader not running"	Supervisor service stopped, or the scale is not Enabled.
"already uses COM3" when saving	Each scale/gate relay needs its own serial port.
Camera test: "No JPEG from the camera URL"	Wrong snapshot URL/credentials, camera unreachable, `curl` extension disabled.
Plate always UNREAD / NOIMG	Camera or service problem (use the test buttons), poor picture, confidence threshold too high.
Many MISMATCH flags	Camera angle/lighting; set policy to *Flag*; try another provider.
Gate button: "relay unreachable" / "did not reply"	Wrong IP/port/COM, wiring, slave ID; use *Test OPEN* in Setup → Gates. For Modbus boards that never answer, untick *Require echo reply*.
Log says OK but the barrier does nothing	Wrong relay channel/coil, or the controller needs a longer pulse.
Gate did not auto-close	Supervisor not running, or auto-close is 0.
"No Oracle PHP driver"	Install oci8 + Instant Client (section 9.3); check `php -m` for the same PHP the services use.
`ORA-12541` / `ORA-12514`	Listener host/port or **service name** wrong.
`ORA-00942`	Table missing or no rights: run `sql/oracle_schema.sql`, grant rights.
Tickets stay PENDING	Oracle sync service not running or Oracle unreachable — see *Diagnostics*.
Upload says the file is too large	Raise `upload_max_filesize` (10M+) and `post_max_size` (12M+) in `php.ini`, restart the web server / PHP.
"OCR is switched off" / "Could not read"	*Setup → Documents & OCR*: choose a provider and use *Read the sample*. Claude: check the API key, model name and internet. Tesseract: check it is installed and the command path.
Everything reads but units are empty	Normal for Tesseract on some layouts – choose the unit on each line (the Exceptions box suggests it). Use the Claude reader for photos.
Document says NO_TICKET but the truck was weighed	Check vehicle number and date on the document; the ticket must be INWARD (invoice) / OUTWARD (return) and inside the date window; press *Link* under *Possible tickets*.
QTY_MISMATCH on every document	Check the unit (MT vs KG) and the tolerance; service lines are ignored, but a wrong unit is not.
Document stays HELD	It has open HIGH exceptions: resolve them with a note, or an administrator ticks *send anyway*.
SAP: "login refused" / "CSRF token validation failed" / "endpoint not found"	Check user/password or token, SAP client, the endpoint URL, and tick *CSRF handshake* for SAP Gateway services.
"Too many failed logins"	Wait 10 minutes, or ask another administrator.
Screens load but look unstyled / links 404	Web server is not pointing at the `public` folder.

When you ask for support, please send: the output of `php bin/check.php`, a screenshot of *Setup → Diagnostics*, the last lines of `data/logs/scale_<n>.log`, the exact message on screen, and the indicator model with its serial output settings.

17. Odoo integration and the REST API

The program keeps its **own local SQLite database** and works fully without any ERP. Odoo (or any other system) is connected in two independent ways; use either or both.

Way	Direction	Use when
A. Push to Odoo (Setup → *Odoo & API* → *Odoo connection*)	Weighbridge → Odoo. Every completed or cancelled ticket is written to the Odoo model `wb.ticket`. Optionally the program also **fetches confirmed purchase orders, vendors/customers and products** from Odoo.	Odoo is in the cloud (Odoo.sh / Online) or anywhere the weighbridge PC can reach but Odoo cannot reach the PC. **Recommended.**
B. REST API (`/api/v1/…`)	Any system → Weighbridge. Read tickets and documents; write purchase orders and master data.	Odoo (or another system) can reach the weighbridge server, or you build your own integration. The Odoo addon includes a ready *pull* job for this.

Both are *store-and-forward*: if Odoo is switched off or the network is down, tickets stay **PENDING** in the local database and are delivered automatically when it returns. Sending the same ticket again updates the same Odoo record (matched on ticket number) – no duplicates. Weighing never waits for Odoo.

17.1 Install the Odoo addon

1. Copy the folder `odoo/wb_weighbridge` into your Odoo **addons path** (on Odoo.sh: into your repository; on Odoo Online custom addons are not possible – use way B with a consultant, or Odoo.sh).
2. Restart Odoo, switch on *Developer mode*, *Apps → Update Apps List*, search **Weighbridge**, *Install*. The addon is written for **Odoo 17** (it depends on `base`, `mail`, `product`, `purchase`). For Odoo 16 or 18 your Odoo partner may need to make small changes to the view tags (`tree`/`list`) and the cron definition.
3. A new menu **Weighbridge → Tickets** appears (list, form, pivot; filters for inward, outward, cancelled, *no purchase order*, plate mismatch). Two groups are created: *Weighbridge / User* (read) and *Weighbridge / Manager* (full); the administrator is a manager.
4. The addon links each ticket to a **partner**, **product** and **purchase order** when the typed name or PO number matches **exactly one** Odoo record; otherwise leave it for a person to choose – a manual choice is never overwritten.

17.2 Connect the weighbridge to Odoo (push)

1. In Odoo create a technical user (or use an existing one) with the *Weighbridge / Manager* group and create an **API key**: *Preferences → Account Security → New API Key*.
2. In the program: *Setup → Odoo & API*. Enter **Odoo URL** (for example `https://mycompany.odoo.com`), **Database**, **Login** and the **API key**. Leave *Odoo model* as `wb.ticket`.
3. Press **Test Odoo connection**. It must say *Connected OK (Odoo 17.0, user id …). Model wb.ticket has N record(s).* If it says the model is not usable, the addon is not installed in that database.
4. Tick **Send completed tickets to Odoo automatically** and **Save**. From now on every ticket you close is queued.
5. Start the sender: *Windows* – the commented lines at the end of `deploy\windows-install.bat` (NSSM service `WeighbridgeOdooSync`); *Linux* – `deploy/weighbridge-odoo-sync.service` (systemd), or cron `* * * * * php /var/www/weighbridge/bin/sync_odoo.php`. **Send pending now** in Setup does the same once by hand.
6. Optional: tick *fetch confirmed purchase orders* and/or *fetch vendors/customers and products*. The sender then imports every confirmed purchase order line (Odoo `purchase.order.line`) into the PO list used by the invoice checks (chapter 13) and adds new vendor / product names to the drop-down lists. Press **Fetch POs / masters now** to try it. Nothing is deleted from the local lists.

Times are converted from the PC's local time to UTC (as Odoo stores them). Check the time zone of the PC.

What is written to `wb.ticket`: ticket_no, state (CLOSED / CANCELLED), direction, vehicle_no, driver, challan_no, party_name, material_name, gross_kg, tare_kg, net_kg, first_weight_at, second_weight_at, operator, scale_name, plate_in, plate_out, plate_flag, remarks, local_id.

Monitoring. *Reports* shows the status per ticket; *Setup → Diagnostics* shows the number of pending/failed Odoo tickets; a failed ticket shows Odoo's own error message and is retried automatically.

17.3 The REST API

Switch it on with *Setup → Odoo & API → REST API → Generate new token* (shown **once**, stored only as a hash). Every request carries the header `Authorization: Bearer <token>`; always use **HTTPS** outside the plant LAN. Reading is always possible with a valid token; **writing** needs the extra switch *Allow API writes*. After 20 wrong tokens from one address the API locks for 5 minutes. A machine-readable description (OpenAPI 3) is at `/api/v1/openapi.php` (no token needed – it contains no data); import it into Postman or Swagger.

Endpoint	Method	What it does
`ping.php`	GET	Checks the token; shows version and whether writes are enabled
`tickets.php?since_id=N&limit=L`	GET	Completed and cancelled tickets with id > N (max 500). The answer carries `next_since_id` – use it as the next `since_id`. Filters: `status`, `from`, `to` (dates), `vehicle`, `party`, `material`. `ticket_no=…` returns one ticket of any status.
`documents.php`	GET	Invoice / return documents with match status and exception counts (filters `since_id`, `status`, `match`, `type`); `doc_no=…` returns one document in full – the same JSON that is sent to SAP (Appendix G) including lines, linked tickets and exceptions
`purchase_orders.php?po=…`	GET	Lines of one purchase order
`purchase_orders.php`	POST	`{"pos":[{"po_no":"P00012","vendor":"ABC","po_date":"2026-09-01","lines":[{"line_no":10,"material_code":"IRON","description":"Iron ore","qty":40,"uom":"MT","rate":1200}]}]}` – replaces all lines of each PO sent (up to 500 POs). If one entry is invalid **nothing** is stored (HTTP 422 with the reason).
`masters.php?type=parties|materials|vehicles`	GET	The lists the operator picks from
`masters.php`	POST	`{"type":"parties","items":[{"name":"ABC Traders","active":true}]}` or `{"type":"vehicles","items":[{"reg_no":"MH31AB1234","tare_kg":9000,"transporter":"XYZ","blocked":false}]}` – upsert; all-or-nothing

Answers are JSON. Errors: `{"error":"…"}` with HTTP 400 (bad request), 401 (token), 403 (writes off), 404, 405 (wrong method), 413, 422 (validation), 429 (locked).

Example:

```
curl -H "Authorization: Bearer $TOKEN" "https://wb.example.com/api/v1/tickets.php?since_id=0&limit=100"
```

17.4 Pull mode (Odoo reads the weighbridge)

If Odoo can reach the weighbridge server: in Odoo set the *System Parameters* `wb.api_url` (for example `https://wb.example.com/api/v1`) and `wb.api_token` (and `wb.api_tz` if the PC is not on `Asia/Kolkata`), then activate the scheduled action **Weighbridge: fetch new tickets** (every 5 minutes), or run *Weighbridge → Fetch now*. The job remembers the last id (`wb.last_id`) and creates or updates `wb.ticket` records by ticket number, so it can run together with the push without duplicates.

17.5 What has and has not been tested

The push client, the PO and master pulls, store-and-forward and the whole REST API are covered by the automated tests against a **simulated Odoo JSON-RPC server** and a running copy of the API (about 65 checks). The Odoo addon was checked for syntax only; it has **not been installed in a live Odoo** here. Install it on an Odoo test database first, run *Test Odoo connection*, close one ticket on a test weighbridge and confirm the record appears, before using it for real. Receiving goods or creating invoices in Odoo from a ticket is not automatic – the tickets are linked to purchase orders so that your Odoo team can build the receipt / bill process they want on top.

18. Appendices

Appendix A — Settings reference

Setting	Where	Default
Company name / address / ticket prefix	General	`My Weighbridge` / — / `WB`
Allow manual weight	General	off
Connection type, port, baud, data bits, parity, stop bits	Scales	Simulator; 9600, 8, N, 1
Protocol, terminator	Scales	ASCII; `\r\n`
Weight regex	Scales	`/([+-]?\d+(?:\.\d+)?)\s*(kg|g|t)?/i`
Stable regex / motion regex	Scales	empty (use repeated readings)
Same reading N times / tolerance	Scales	5 / 0 kg
Divisor	Scales	1
Minimum weight	Scales	20 kg
Camera URL, user, password	Scales, Gates	empty
Gate driver / mode / pulse / auto-close / automatic opening	Gates	Simulator / pulse / 1000 ms / 0 / on
ANPR provider, URL, token, region	Plate recognition	off
ANPR minimum confidence / policy / auto-read	Plate recognition	0.6 / flag / off
Weighing requires gate entry / exit rule	Gates	off / warn
Oracle host, port, service, user, password, table, batch	Oracle transfer	— / 1521 / — / — / — / `WEIGHBRIDGE_TICKETS` / 50
Invoice reader, API key, model, effort	Documents & OCR	off; —; `claude-opus-5-5`; medium
Tesseract command	Documents & OCR	`tesseract {image} stdout -l eng --psm 6`
Max upload / date order	Documents & OCR	10 MB / day-first
Date window / quantity tolerance	Documents & OCR	5 days / 2 %
"Weighed, no invoice": look back / grace	Documents & OCR	14 days / 24 hours
Send documents to / hold policy	Documents & OCR	nowhere / hold if HIGH exceptions
Oracle document tables	Documents & OCR	`WB_DOCUMENTS`, `WB_DOCUMENT_LINES`
SAP endpoint, login, client, CSRF, reference path, PO lookup, timeout	Documents & OCR	empty; basic; —; on; —; —; 15 s
Odoo on/off, URL, database, login, API key, model, timeout, batch	Odoo & API	off; —; —; —; —; `wb.ticket`; 20 s; 50
Fetch Odoo purchase orders / vendors & products	Odoo & API	off / off
REST API token / allow API writes	Odoo & API	none / off

Appendix B — Oracle table `WEIGHBRIDGE_TICKETS`

Column	Type	Content
`TICKET_NO`	VARCHAR2(30), primary key	Ticket number, e.g. `WB202609290001`
`VEHICLE_NO`	VARCHAR2(30)	Vehicle registration
`PARTY`, `MATERIAL`, `DIRECTION`, `DRIVER`, `CHALLAN_NO`, `REMARKS`	VARCHAR2	As entered (`DIRECTION` = INWARD / OUTWARD)

Column	Type	Content
`GROSS_KG`, `TARE_KG`, `NET_KG`	NUMBER(12,2)	Weights in kg
`FIRST_WEIGHT_AT`, `SECOND_WEIGHT_AT`	TIMESTAMP	Local time of each weighment
`STATUS`	VARCHAR2(12)	`CLOSED` or `CANCELLED`
`OPERATOR`	VARCHAR2(50)	User who created the ticket
`LOCAL_ID`	NUMBER(12)	Record number in the weighbridge database
`SCALE_NAME`	VARCHAR2(100)	Scale used for the first weighment
`PLATE_IN`, `PLATE_OUT`, `PLATE_FLAG`	VARCHAR2	Plates read, and OK / MISMATCH / UNREAD / NOIMG
`SYNCED_AT`	TIMESTAMP	When the row was written

Appendix B2 — Oracle tables `WB_DOCUMENTS` and `WB_DOCUMENT_LINES`

Column (`WB_DOCUMENTS`)	Content
`DOC_NO` (primary key)	Document number, e.g. `DOC202609290001`
`DOC_TYPE`	`PO_INVOICE` or `MATERIAL_RETURN`
`INVOICE_NO`, `INVOICE_DATE`	Number and date on the document
`SUPPLIER`, `SUPPLIER_TAX_ID`, `BUYER`	Parties as verified
`PO_NO`, `REF_NO`, `ORIG_INVOICE_NO`	PO number; return reference; original invoice (returns)
`VEHICLE_NO`, `EWAY_NO`, `CURRENCY`	Truck, e-way bill / LR, currency
`SUBTOTAL`, `TAX_AMOUNT`, `TOTAL_AMOUNT`	Amounts
`MATCH_STATUS`	`MATCHED`, `REVIEW`, `EXCEPTION`, `UNMATCHED`
`WB_TICKETS`, `WB_NET_KG`	Linked weighbridge ticket numbers; their total net weight
`EXC_HIGH`, `EXC_WARN`, `EXC_SUMMARY`	Number of open HIGH / warning exceptions; codes of the open ones
`VERIFIED_BY`, `VERIFIED_AT`, `SYNCED_AT`	Who verified, when, when written

`WB_DOCUMENT_LINES` (key `DOC_NO`, `LINE_NO`): `MATERIAL_CODE, DESCRIPTION, HSN, QTY, UOM, RATE, AMOUNT, QTY_KG`.

Appendix C — Indicator output examples

These are only **examples of the kind of line** indicators send. Always check your indicator's manual and use *Test connection* to see what yours really sends.

Line received (each ends with CR LF)	Works with the default regex?	Note
`ST,GS,+0012340kg`	✔	Enter Stable regex `/\bST\b/` and Motion regex `/\bUS\b/` if the line begins with ST/US
`12340 kg`	✔	
`=0012340`	✔	Unit defaults to kg
`012340` meaning 1234.0 kg	✔ number, ✘ scale	Set **Divisor** to 10
`W 1.5 t`	✔	Converted to 1500 kg
Frame `<STX> 5000 kg <ETX>`	✔	Terminator `\x03`

Appendix D — Commissioning and acceptance checklist

Site		**Date**	
Engineer		**Client witness**	

#	Test	Result
1	`php bin/check.php` shows no `[FAIL]`	☐

#	Test	Result
2	Installer completed; administrator and operator logins work	☐
3	Services start automatically after a PC restart (readers, web, Oracle sync)	☐
4	Each scale displays the correct weight; empty bridge reads 0 (or below minimum)	☐
5	Known test weight placed on each bridge reads correctly	☐
6	STABLE appears only when the weight is really steady; buttons are disabled during MOTION	☐
7	Two-pass ticket: gross, tare and net are correct; slip prints	☐
8	Single-pass ticket with stored tare is correct	☐
9	Second weighment on a different bridge works	☐
10	Unplugging the indicator shows NO DATA within 3 s and blocks capture	☐
11	Camera test returns a picture; photo appears on the slip	☐
12	Plate recognition reads at least 8 of 10 real vehicles (or is left off)	☐
13	Each gate: Test OPEN and Test CLOSE move the barrier (people clear); barrier's safety devices verified	☐
14	Gate entry → weighing → exit works; exit refused with an open ticket (if rule is on)	☐
15	Blocked vehicle is refused at the gate	☐
16	Oracle test succeeds; a ticket appears in Oracle with correct values and status SYNCED	☐
17	Oracle down for a while: weighing continues, tickets are sent when it returns	☐
18	Report and CSV export match the slips	☐
19	Backup taken and a test restore checked; `app.key` copied elsewhere	☐
20	Operators trained; administrator password handed over securely	☐
21	`php bin/check.php` shows upload limits ≥ 10M/12M, and Tesseract or the Anthropic key works (*Read the sample* on a real invoice)	☐
22	Ten real invoices read; every field compared with the picture; units and totals correct or corrected	☐
23	A purchase invoice for a weighed truck matches automatically; quantity difference within tolerance shows MATCHED	☐
24	A wrong quantity, a wrong vehicle and an unweighed truck each raise the expected exception; resolving with a note clears it	☐
25	A material return matches an OUTWARD ticket and its original invoice; over-return is caught	☐
26	A verified document reaches SAP (and/or the Oracle tables) in the test system with the right values; a held document is not sent until released	☐
27	Odoo (if used): addon installed on a test database; *Test Odoo connection* says Connected OK; a closed ticket appears in *Weighbridge → Tickets* with the right weights and times	☐
28	Odoo switched off for a while: weighing continues, tickets show PENDING/FAILED and arrive after it returns, without duplicates; a cancelled ticket updates the same record	☐
29	REST API: token generated, `ping.php` answers, a wrong token is refused, `tickets.php` returns the new ticket; writes stay refused until *Allow API writes* is ticked	☐

Appendix E — Main database tables (SQLite)

`weighments` (tickets), `gate_entries` (gate passes), `documents`, `document_lines`, `document_tickets` (links to tickets), `doc_exceptions`, `po_lines` (PO list), `vehicles`, `parties`, `materials`, `scales`, `live_scale` (current readings), `gates`, `gate_state`, `gate_events`, `users`, `settings`, `audit`, `login_attempts`. Tickets carry `odoo_status`, `odoo_id`, `odoo_at`, `odoo_error` (state of the Odoo transfer). The file is `data/weighbridge.sqlite`; it can be opened read-only with any SQLite tool for your own reports. Please do not edit it while the system is running.

Appendix F — Glossary

Term	Meaning
Gross / Tare / Net	Loaded weight / empty weight / material weight (Gross − Tare).
Indicator	The display unit connected to the load cells; it sends the weight to the PC.

Term	Meaning
RS-232 / RS-485	Serial connections. RS-485 uses two wires and works over long distances and multiple devices.
Modbus	A common industrial protocol used by transmitters, relays and PLCs.
Stable	Weight is steady (no motion), safe to record.
ANPR	Automatic number-plate recognition.
Boom barrier	The gate arm at the entry, exit or weighbridge.
OCR	Optical character recognition – turning a picture of text into data.
PO	Purchase order. A purchase invoice should refer to one.
Exception	A difference between a document and the weighbridge (or the PO) that a person must look at.
Verified	A person has checked the document against the picture and locked it.
Pulse	A short contact closure, like pressing a remote-control button.
Reader / supervisor	Background programs that read a scale / that keep all readers running.
Store-and-forward	Records are kept locally and sent when the destination is available.

Appendix G — Data sent to SAP (JSON contract)

One HTTP `POST` per document, `Content-Type: application/json`. Fields that are not known are `null`. The example is a real document from the test system.

```
{
    "documentNo": "DOC202609290001",
    "documentType": "PO_INVOICE",
    "invoiceNo": "BM/2026-27/0451",
    "invoiceDate": "2026-09-15",
    "supplier": "SHREE BALAJI MINERALS PVT LTD",
    "supplierTaxId": "27AABCS1234F1Z5",
    "buyer": "Sunrise Steel Pvt Ltd",
    "poNo": "4500012345",
    "referenceNo": null,
    "originalInvoiceNo": null,
    "vehicleNo": "MH31AB1234",
    "ewayBillNo": "481234567890",
    "currency": "INR",
    "subtotal": 139650,
    "taxAmount": 25137,
    "totalAmount": 164787,
    "lines": [
        {
            "lineNo": 1,
            "materialCode": null,
            "description": "Iron Ore Fines 62% Fe",
            "hsn": "2601",
            "quantity": 28.5,
            "uom": "MT",
            "rate": 4850,
            "amount": 138225,
            "quantityKg": 28500
        },
        {
            "lineNo": 2,
            "materialCode": null,
            "description": "Loading and Handling Charges",
            "hsn": "9967",
            "quantity": 28.5,
            "uom": "MT",
            "rate": 50,
            "amount": 1425,
            "quantityKg": 28500
        }
    ],
    "weighbridge": {
        "ticketNos": [
            "202609150001"
        ],
        "netKg": 28450,
        "tickets": [
            {
                "ticket_no": "202609150001",
                "vehicle_no": "MH31AB1234",
                "direction": "INWARD",
                "net_kg": 28450,
                "status": "CLOSED"
            }
        ]
    },
    "matchStatus": "MATCHED",
    "exceptions": [],
    "verifiedBy": "admin",
    "verifiedAt": "2026-09-29 14:21:22"
}
```


Field	Meaning
`documentNo`	Unique number of the document in this system – use it as an idempotency key (the same document may be posted again with changes)

Field	Meaning
`documentType`	`PO_INVOICE` or `MATERIAL_RETURN`
`invoiceNo`, `invoiceDate`	As verified (date `YYYY-MM-DD`)
`poNo`, `referenceNo`, `originalInvoiceNo`	PO number (invoices); return reference and original invoice (returns)
`lines[]`	`lineNo, materialCode, description, hsn, quantity, uom, rate, amount, quantityKg` (`quantityKg` only for weight units)
`weighbridge`	Linked tickets and their total net weight in kg
`matchStatus`	`MATCHED`, `REVIEW`, `EXCEPTION`, `UNMATCHED`
`exceptions[]`	`code, severity, message, status` (`OPEN`, `RESOLVED`, `WAIVED`) and the reviewer's `note`
`verifiedBy`, `verifiedAt`	Who locked the document, and when

Any 2xx answer means "accepted". If the answer is JSON, the SAP document number is read from the path configured in Setup (for example `d.MaterialDocument`), or else from `documentNo`, `DocumentNumber`, `MaterialDocument`, `id` or `ref` if present. A non-2xx answer marks the document FAILED with the message SAP returned (for OData errors `error.message.value`), and it is retried.

Optional PO lookup (SAP → program). A `GET` to the configured URL with `{po}` replaced by the PO number. Accepted answers: `{"po_no":"…","vendor":"…","lines":[{"line_no":10,"material_code":"…","description":"…","qty":40,"uom":"MT","rate":1200}]}`, or the standard OData shapes (`d.to_PurchaseOrderItem.results[]`, or OData v4 `value[]` with `_PurchaseOrderItem`) using the fields `PurchaseOrder, Supplier, PurchaseOrderItem, Material, PurchaseOrderItemText, OrderQuantity, PurchaseOrderQuantityUnit, NetPriceAmount`. HTTP 404 means "PO not found".

Appendix H — Exception catalogue

Code	Sev.	Meaning	What to do
`NO_TICKET`	HIGH	No weighbridge ticket found for the document	Was the truck weighed? Check vehicle number and date; link a ticket by hand; or resolve/waive with a note (weighed elsewhere)
`QTY_MISMATCH`	HIGH	Invoice quantity differs from the weighbridge net weight beyond the tolerance	Check unit and quantity on the picture; talk to the supplier; resolve with the agreed outcome
`VEHICLE_MISMATCH`	HIGH / WARN	Vehicle on the document differs from the ticket (WARN when only one character differs)	Correct a misread number, or find out which truck it was
`DIRECTION_MISMATCH`	HIGH	Purchase invoice linked to an OUTWARD ticket, or a return to an INWARD ticket	Unlink and link the right ticket
`TICKET_CANCELLED`	HIGH	The linked ticket was cancelled	Link the correct ticket
`TICKET_MULTI_DOC`	HIGH	The same ticket is linked to two documents (possible double billing)	Decide which document owns the ticket; unlink the other
`DUPLICATE_INVOICE`	HIGH	Same invoice number from the same supplier was uploaded before	Cancel the duplicate (administrator) or resolve with a note
`PO_MISSING`	HIGH	Purchase invoice without a PO number	Enter the PO number from the picture / ask the supplier
`PO_UNKNOWN`	HIGH	PO not in the PO list / SAP	Import the PO list; check the number
`PO_VENDOR_MISMATCH`	HIGH	Supplier is not the vendor of the PO	Check the supplier name; wrong PO?
`PO_OVER_QTY`	HIGH	Total invoiced on a PO line exceeds the ordered quantity	Check other invoices on the same PO; get a PO amendment
`RETURN_OVER_QTY`	HIGH	Returned quantity (all returns together) exceeds the quantity invoiced	Check the return quantities
`PARTY_MISMATCH`	WARN	Supplier differs from the party on the ticket	Correct the party name on either side
`MATERIAL_MISMATCH`	WARN	Ticket material does not resemble any invoice line	Check the material

Code	Sev.	Meaning	What to do
`DATE_MISMATCH`	WARN	Ticket date is further from the document date than the date window	Check the dates
`TICKET_OPEN`	WARN	Linked ticket has no second weighment yet	Complete the ticket
`UOM_MISSING`	WARN	A line with a quantity has no unit (a suggestion is shown)	Choose the unit
`NO_INVOICE_NO`, `NO_SUPPLIER`, `NO_DATE`, `NO_LINES`	WARN	Basic details missing	Fill them in
`PO_LINE_UNKNOWN`	WARN	An invoiced item is not on the PO	Check the item / PO
`PO_RATE_MISMATCH`	WARN	Rate differs from the PO rate by more than 1 %	Check the rate
`RETURN_NO_REF`	WARN	The return does not name an original invoice	Enter it
`RETURN_ORIG_UNKNOWN`	WARN	The original invoice is not among the uploaded documents	Upload it
`RETURN_SUPPLIER_MISMATCH`	WARN	Return is to a different supplier than the invoice	Check
`RETURN_LINE_UNKNOWN`	WARN	Returned item is not on the original invoice	Check
`NO_DOCUMENT`	WARN	An inward ticket has no invoice document (listed on the Exceptions screen)	Upload the invoice or link the ticket
`QTY_UNCHECKED`	INFO	Quantity could not be compared (units are pieces, litres…)	None
`PO_NOT_CHECKED`	INFO	No PO list / SAP lookup available	Load the PO list if you want PO checks

---

End of manual — Weighbridge Management System v1.1.0